The Geometric Architecture of Matter and the Cosmos

General Relativity, Geometric Mass, and the Death of the Dark Matter Paradigm

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Abstract

Gravitational mass is inferred from the response of a gravitational field; material mass is reconstructed from the matter that sources it. Curvature Polarization Transport Gravity (CPTG) develops the physical distinction between these quantities into a unified geometric framework. The Einstein tensor and ordinary material stress–energy remain continuously present, while curvature polarization and transport contribute an organized gravitational response. Geometric mass expresses that response without adding an unseen material inventory.

The theory's physical meaning is developed from its parent source–curvature equation to the different forms in which organized geometry is observed. In rotating galaxies the completed field admits an acceleration-equivalent mass accounting distinct from the independently reconstructed stellar and gas inventory. In clusters and diffuse stellar systems pressure and random motions provide different observational support for the same gravitational architecture. At cosmic scale a fixed native curvature–transport history yields acceleration without a separate dark-energy fluid, while acoustic and luminosity measurements occupy explicitly projected comparison coordinates. The light-element nuclear program has its own conserved reaction-source geometry and a distinct mass-seven transport prescription. In compact objects a finite interior and fixed mass–spin handoff separate localized source mass from conserved exterior gravitational charge, with consequences for singular endpoints, compact mass gaps and the geometry of time. Structural Mode N measures organization only after the gravitational field has been completed. Each physical regime has its own defined observables and reductions, while the underlying geometric organization remains the unifying principle.

Together these results establish a concrete alternative way to interpret gravitational mass discrepancies: the field can express more gravity than the directly inventoried material source, because gravitational response also depends on geometric organization.

1 Why the meaning of gravitational mass matters

A galaxy's gravitational field and material inventory are reconstructed from complementary observations. Stellar light and gas trace the amount and distribution of ordinary matter; rotational and random motions describe the gravitational field in which that matter moves. The dynamical mass equivalent calculated from the field is therefore a physical quantity distinct from the material mass reconstructed from the source. The prevailing ΛCDM interpretation attributes their difference to an additional cold-dark-matter component. Curvature Polarization Transport Gravity (CPTG) develops a geometric account of the same measured gravitational response.[4, 3, 1]

For example, an observed rotation speed specifies the strength of a galaxy's circular gravitational field at a given radius, whereas measured starlight and gas constrain how much ordinary matter is present and where it lies. CPTG reconstructs the coupled field from this material and the observed dynamical support. A spherical-reference mass equivalent of the completed field can then be compared with the separately integrated, generally flattened source inventory. Their difference is a defined gravitational accounting quantity, not a census of newly discovered particles. The compact-object mass split is a different, conserved exterior-charge construction.[4, 7]

In CPTG, material matter and its motion remain explicit, Einstein curvature remains continuously present, and curvature polarization and transport contribute to the organized field response. The governing source and support conditions connect that response to the observed spatial and dynamical structure. The same architecture connects galaxies, clusters, cosmological comparison coordinates, primordial nuclear reactions, and finite compact systems, while distinguishing field-derived gravitational equivalents from conserved compact charge.[2, 3, 4, 7]

1.1 One gravitational framework with the Einstein term continuously present

At the gravitational parent level, the framework is expressed through the variational architecture[2]

$$S_{\rm CPTG} = \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\mathcal{R} + \mathcal{L}_{\rm pol} + \mathcal{L}_{\rm trans} + \mathcal{L}_{\rm m}\right], \tag{1}$$

where $\mathcal{R}$ is the Ricci scalar, $\mathcal{L}_{\rm m}$ describes ordinary material stress–energy, and the two geometric contributions organize polarization and transport. Equation (1) states the common *gravitational* action structure; explicit expressions used for different physical configurations are registered reductions, not one scalar coordinate formula applied indiscriminately at all scales.[2]

The extra response tensors are defined by metric variation of their action contributions. Writing $S_{\rm pol} = \int d^4x\sqrt{-g}\,\mathcal{L}_{\rm pol}$ and similarly for $S_{\rm trans}$, the convention of the parent paper gives[2]

$$\Pi^{\rm pol}_{\mu\nu} = \frac{16\pi G}{c^4\sqrt{-g}}\,\frac{\delta S_{\rm pol}}{\delta g^{\mu\nu}}, \qquad\qquad \Theta^{\rm trans}_{\mu\nu} = \frac{16\pi G}{c^4\sqrt{-g}}\,\frac{\delta S_{\rm trans}}{\delta g^{\mu\nu}}. \tag{2}$$

With the usual material stress–energy convention, the parent equation is

$$G_{\mu\nu} + \Pi^{\rm pol}_{\mu\nu} + \Theta^{\rm trans}_{\mu\nu} = \frac{8\pi G}{c^4}T^{\rm material}_{\mu\nu}. \tag{3}$$

The same equation can be rearranged without introducing any new physical source. Define an *equivalent geometric response tensor* by

$$T^{\rm geom,eq}_{\mu\nu} \equiv -\frac{c^4}{8\pi G}\big(\Pi^{\rm pol}_{\mu\nu} + \Theta^{\rm trans}_{\mu\nu}\big), \qquad G_{\mu\nu} = \frac{8\pi G}{c^4}\big(T^{\rm material}_{\mu\nu} + T^{\rm geom,eq}_{\mu\nu}\big). \tag{4}$$

This is an exact algebraic identity within the registered parent field equation, not a second material stress–energy tensor. It shows why a geometrically organized response can be *expressed* as a gravitating equivalent without being *identified* with an inventory of particles. The tensor-level identity, the galaxy's radius-by-radius spherical mass equivalent, and the conserved compact exterior-charge split are related interpretations at different levels of description, not identical density measurements.[2, 4, 7]

Neither the Einstein tensor nor material stress–energy is switched off. The added response is geometric: $\Pi^{\rm pol}_{\mu\nu}$ and $\Theta^{\rm trans}_{\mu\nu}$ are not postulated densities of invisible matter. Where their *combined* contribution becomes negligible, the same field

equation approaches the Einstein equation for the same source,[2]

$$\Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \longrightarrow 0 \quad \Longrightarrow \quad G_{\mu\nu} = \frac{8\pi G}{c^4} T^{\mathrm{material}}_{\mu\nu}. \tag{5}$$

The Einstein equation is recovered continuously within the same coupled framework, with the magnitude of the geometric response determined by the relevant physical state.

1.2 Conservation and the status of geometric directions

The contracted Bianchi identity provides a non-negotiable consistency condition. If the ordinary material tensor is independently conserved, $\nabla^\mu T^{\mathrm{material}}_{\mu\nu} = 0$, then Eq. (3) requires[2]

$$\nabla^\mu \left(\Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \right) = 0. \tag{6}$$

The condition governs the *sum*; neither geometric tensor is required to be independently divergence-free. In particular, the parent action does not authorize an arbitrary spacetime-dependent transport direction prescribed without dynamical justification. If a geometric direction or state variable is independent, its Euler–Lagrange equation must be satisfied; alternatively it must be obtained covariantly from the relevant solved state. A direction may be held fixed within an explicitly declared reduction after that reduction has selected it.[2]

The parent also fixes how apparently different solutions can belong to one framework. If two reductions A and B possess a physical overlap domain $\mathcal{O}$, then each quantity with the *same physical meaning* in both reductions must agree there:[2]

$$Q_A\big|_{\mathcal{O}} = Q_B\big|_{\mathcal{O}}. \tag{7}$$

A rotating disk, spherical cluster and finite compact configuration need not use identical component equations. Their common status depends on the fixed source–response architecture, valid reduction assumptions, conservation, and agreement in real overlap regions.

1.3 Why motion and spatial organization are part of the state

The physical state contains more information than total material mass. The spatial source distribution matters, as do the velocity and support structure through which gravity is observed and reconstructed. Different kinematic organizations select different admissible reductions and boundary data: ordered disk rotation; stellar velocity dispersion; hydrostatic gas pressure; or relativistic compact mass, spin and geometry. This does not create a separate fundamental gravity law for each object. It specifies the physical problem to which the common architecture is applied.[3, 2]

For the disk row of the continuity ledger below, the local structural acceleration is evaluated from the observed support at each measured radius, and the smooth reconstructed support profile supplies gradients used by the polarization and transport operators. The published radial field implementation also recovers object-specific reduced response coordinates from observational closure. Those quantities describe the adopted reduction and its fitted realization; they must be reported as such rather than described as newly added material.[3, 4]

1.4 How one gravitational architecture appears in different regimes

The parent equation, Eq. (3), and its combined conservation condition, Eq. (6), are the common dynamical and geometric contract. Solving a source with different symmetry and support imposes different reduction conditions, not a new material component. Table 1 makes the inheritance and matching requirements explicit; the sector operators are those *registered in their source studies*, not four newly derived field equations.[2, 3, 7, 14]

State	Parent-law and source inheritance	Reduced solution / overlap	Structural status
Resolved disk	Einstein plus polarization/transport; signed stellar/gas distribution, measured rotation and radial support	Measured-support radial field and local derivatives; continuous weak-field GR overlap	Local $a_\star$ and object response inferred from support
Cluster/stellar system	Same tensor architecture; ordinary matter, gas density/temperature/pressure or stellar velocity moments	Hydrostatic or Jeans observation operator; tensor geometry where needed	Support and covariance chosen per observation
Cosmic background	Native curvature–transport branch; fixed $p_C = \pi$, $B_\pi + T_\pi = 1$	$E_\pi(a)$ and downstream acoustic/luminosity projections	Registered geometric background, no fluid-fit parameter
Compact state	Same Einstein exterior; localized source, exterior mass, spin and relativistic geometry	Finite spherical profile, exterior matching and conditional spin handoff	Exterior input (M_∞, q); N measured after solution

Table 1: Checkable continuity ledger of the registered gravitational realizations. A shared gravitational architecture does not mean that distinct symmetry reductions can be substituted for one another. Nuclear reaction-space currents obey their own baryon-conservation identity rather than the gravitational Einstein equation.

The ledger tracks the inherited Einstein and material terms separately from each regime's solution and observational interface. In the disk, the registered signed source and measured support supply a radial field root; in pressure systems, the same general source–response architecture is accessed through hydrostatic or Jeans support; the cosmological reduction specifies a native history and then defined projection clocks; the compact realization uses a finite profile and an exterior matching condition. It does not identify these different reduced equations term by term. At an overlap boundary the matching condition $Q_A|_{\mathcal{O}} = Q_B|_{\mathcal{O}}$ applies to the same physical quantity in both descriptions. In the compact example the exterior charge agrees at the finite matching surface; in the weak-response limit the tensor sum approaches zero and the ordinary Einstein equation is recovered. The cosmological branch instead specifies the native global background and then projects it into observed comparison coordinates. The comparison distinguishes what is shared at the parent-law level from what each registered solution must supply.[2, 7, 14]

1.5 Polarization, transport, and observational projection are distinct

The parent distinguishes three operations. *Polarization* describes the nonlinear organization of sourced curvature. *Transport* describes its directional persistence and redistribution through an extended or changing state. Only after that physical state has been solved does an *observational projection* express it in quantities such as rotation, distance, temperature, or time readouts. A projection into observational coordinates is not an additional physical transport force.[2]

The reduced static-field representative illustrates the distinction without claiming a universal scalar formula for every geometry. In an appropriate weak-field reduction,[2, 3]

$$\mathcal{C}_{\rm red} = g_j g_j + \kappa_{\rm red}^2 (\partial_i g_j)(\partial_i g_j), \tag{8}$$

$$\mathcal{L}_{\rm pol}^{(\rm red)} = \beta a_\star^{2/3} \mathcal{C}_{\rm red}^{1/3}, \qquad \mathcal{L}_{\rm trans}^{(\rm red)} = -\frac{\lambda_t}{2} \left(S^\alpha \Sigma_{\rm red} \nabla_\alpha \Phi\right)^2 . \tag{9}$$

Here $\kappa_{\rm red}$ is the declared gradient length, $a_\star$ is the structural acceleration in that reduction, S^α is an admissibly selected transport direction, and $\Sigma_{\rm red}$ describes its reduced response structure. Equation (9) shows how field strength, spatial derivatives and directional organization can enter the geometric terms. The parent text does not assert that this static reduction is valid through black-hole horizons or that its coefficients can be carried unaltered into unrelated geometries.[2]

1.6 The completed field precedes its structural fingerprint

Once the gravitational field is reconstructed or solved in its declared geometry, one may calculate derived observables and gravitational mass equivalents. For example, the galaxy atlas defines a spherical *equivalent* accounting of a resolved radial field; it does not claim that the galaxy's physical material is spherically distributed. A compact object's conserved exterior mass involves a different geometric definition. Neither the masses nor the conventions should be transferred between the two systems without an explicit derivation.[4, 7]

Structural Mode is evaluated still further downstream. Its geometry-dependent definition is given in Section 2 and its distinct measurement operators alongside the galaxy, cluster and compact applications. It is a fingerprint of the *completed* gravitational organization, not a coefficient used to establish the field or an adjustable response multiplier.[5]

These distinctions establish the physical basis for the different regimes. Galaxies and clusters test distinct dynamical organizations; finite compact states test the separation of localized source and exterior gravitational charge; cosmological results use declared projections of a native physical state. The nuclear reaction program shares the source–polarization–transport organization in a conserved reaction-state space while retaining its own native reaction-network equations. The commonality to be demonstrated is physical structure, conservation and cross-regime consistency, not identical numerical formulas where the physics differs.[2, 8]

2 Geometric mass and the organization of gravity

The phrase *geometric mass* requires a definition before it can bear an interpretive claim. In CPTG it does not denote a newly inventoried population of particles, an added material density, or one universal percentage assigned to gravitational systems. The cited studies use two related but physically distinct readouts: a gravitational *mass equivalent* of a completed galactic field relative to an independently reconstructed material inventory, and a conditional separation of conserved exterior gravitational mass from localized compact-source mass. Structural Mode N is a third, separate quantity: an organization measurement obtained *after* the gravitational state is completed. None of these readouts supplies the gravitational solution from which it is obtained.[4, 7, 5]

2.1 A galaxy: the acceleration-equivalent convention

For a rotating galaxy at an observed physical radius R, let $g_{\rm C}(R) > 0$ be the inward circular acceleration returned by the registered CPTG radial field solve and $V_{\rm C}^2(R) = Rg_{\rm C}(R)$ its circular-speed equivalent. The atlas defines the *spherical-reference gravitational mass equivalent*[4]

$$M_{\rm grav}^{\rm eq}(R) \equiv \frac{R^2 g_{\rm C}(R)}{G} = \frac{RV_{\rm C}^2(R)}{G}. \tag{10}$$

This identity answers a limited but meaningful question: what mass, placed in a spherical reference field, would produce the *same circular acceleration at this radius*? It does *not* assert that the observed disk is spherical or that the equivalent is a direct integral of a physical three-dimensional density. The reference-field convention is needed precisely because flattened stellar and gas disks cannot, in general, be converted to enclosed mass merely by multiplying their disk acceleration by R^2/G.

The material inventory is independently assembled from the adopted 3.6-μm stellar photometry, bulge prescription, and signed-gas-force inversion. In the atlas the stellar mass-to-light ratios are fixed for this accounting at $\Upsilon_{\star,\rm disk} = 0.5$ and $\Upsilon_{\star,\rm bulge} = 0.7$, while the gas reconstruction uses nonnegative annular surface densities and its declared regularization. The stellar disk and gas inventory are cylindrical-aperture quantities; the bulge follows its registered spherical approximation. The result is denoted $M_{\rm material}(< R)$ and is conditional on those photometric, geometrical, and gas-inversion assumptions. No dark-material halo is inserted into this inventory. The circular speed used in the field solve is *not* used to select this material mass.[4]

At precisely the same tabulated radius, the atlas then defines[4]

$$M_{\rm geom}^{\rm eq}(R) \equiv M_{\rm grav}^{\rm eq}(R) - M_{\rm material}(< R), \tag{11}$$

$$f_{\rm geom}(R) \equiv \frac{M_{\rm geom}^{\rm eq}(R)}{M_{\rm grav}^{\rm eq}(R)} = 1 - \frac{M_{\rm material}(< R)}{M_{\rm grav}^{\rm eq}(R)}, \tag{12}$$

$$M_{\rm grav}^{\rm eq}(R) = M_{\rm material}(< R) + M_{\rm geom}^{\rm eq}(R). \tag{13}$$

Equation (13) is an exact arithmetic identity within the declared accounting, *not* a second gravitational field equation. The computed $M_{\rm geom}^{\rm eq}$ is therefore distinct from the signed polarization–transport contribution to the *radial acceleration* in the underlying resolved solver. In particular, one cannot identify the independently reconstructed disk material-force curve with $GM_{\rm material}(< R)/R^2$, or add the displayed geometric-equivalent speed linearly to the material-source speed.[4]

2.2 Separating field response from reference-mass bookkeeping

The identity above is not by itself a measurement of the separate polarization and transport terms. In particular, the registered atlas distinguishes the unit-normalized signed source g_b used by the implicit solver, the independently calibrated *material-force display*, and the reconstructed *material inventory*. They are not interchangeable observables. The signed material-force acceleration is, by the atlas convention,[4]

$$g_{\mathrm{mat,force}}(R) = \frac{Q(V_{\mathrm{gas}}) + 0.5V_{\mathrm{disk}}^2 + 0.7V_{\mathrm{bulge}}^2}{R}, \qquad Q(V) = V|V|. \tag{14}$$

The definition permits a locally outward signed gas contribution; $g_{\mathrm{mat,force}}$ is a *radial-force representation*, not a spherical enclosed-mass estimator. Adding and subtracting $R^2 g_{\mathrm{mat,force}}/G$ makes the interpretation of the atlas remainder more precise:

$$M_{\mathrm{geom}}^{\mathrm{eq}}(R) = \underbrace{\frac{R^2}{G}\left[g_{\mathrm{C}}(R) - g_{\mathrm{mat,force}}(R)\right]}_{\Delta M_{\mathrm{field|force}}^{\mathrm{eq}}(R)} + \underbrace{\left[\frac{R^2 g_{\mathrm{mat,force}}(R)}{G} - M_{\mathrm{material}}(<R)\right]}_{\Delta M_{\mathrm{force|inventory}}^{\mathrm{eq}}(R)}. \tag{15}$$

This exact *diagnostic identity* is an algebraic clarification of the published atlas definitions; it is not a newly postulated CPTG field equation. The first term compares the completed circular field to the calibrated material-force curve. Even that difference cannot automatically be identified term for term with polarization or transport, because the registered solver is sourced by the *unit-template* g_b and uses the measured dynamical-support field and object-specific reduced response coefficients. The second term compares a spherical-force reference equivalent with a cylindrical-aperture disk/gas inventory and spherical-reference bulge inventory. It need not vanish even if $g_{\mathrm{C}} = g_{\mathrm{mat,force}}$: flattened geometry, external-ring forces, and reconstruction conventions need not produce the spherical relation $g_{\mathrm{mat,force}} = GM_{\mathrm{material}}(<R)/R^2$. Both terms can be signed.

Thus, the atlas's phrase *geometric mass equivalent* has a precise, useful meaning as the total declared *gravitational-versus-material accounting remainder*; its full numerical value must not be reported as a separately measured positive polarization density or as the unique force contribution of $\Pi_{\mu\nu}^{\mathrm{pol}} + \Theta_{\mu\nu}^{\mathrm{trans}}$. Establishing a source-consistent physical field-channel decomposition is a different calculation from the arithmetic mass equivalent.[4, 2]

2.3 A compact object: conserved exterior gravitational charge

The compact realization makes a related distinction in a different physical setting. Its M_∞ is the exterior gravitational mass parameter retained in the Kerr-compatible description; M_c is the localized contribution assigned to the stationary compact source after formation; and M_{geom} is the geometric mass-equivalent part of the same conserved exterior state:[7]

$$M_\infty = M_c + M_{\mathrm{geom}} = A_{\mathrm{pol}} M_c, \qquad A_{\mathrm{pol}} \equiv \frac{M_\infty}{M_c}. \tag{16}$$

This is *not* obtained by inserting a galactic circular-speed field into Eq. (10). It is the mass accounting of a conditional compact mass–spin handoff that preserves the exterior gravitational mass and total angular momentum. Moreover M_c is not, by definition, the progenitor's mass or a unique microscopic rest-mass integral.[7]

The fixed mass–spin relation, its physical branch, the finite interior channel decomposition, and the two compact examples are derived and displayed together in Section 8. The exterior charge here is not a spherical-reference disk mass equivalent.[7, 2]

2.4 Structural Mode: a measurement taken from the completed field

Neither geometric mass equivalent nor total material mass determines Structural Mode by itself. The mode's common structural coordinate is[5]

$$N \equiv \frac{R}{\lambda}, \tag{17}$$

where R is the declared retained structural extent and λ is the organization length obtained from the *solved* state. The meaning is shared, but λ is measured by the operator appropriate to each geometry. The sequence is *physical state* $\rightarrow$ *solved CPTG field* $\rightarrow$ *Structural Mode*, never the reverse.[5]

The full galaxy estimator, including the published sampled-grid convention, belongs to Section 3; the spherical shell-weighted cluster estimator is developed in Section 4; and the finite compact equal-measure coordinate is developed in Section 8. Each is measured on its own completed field, never used to fit that field.[5, 7]

2.5 The meaning of the proposed unification

A galactic spherical-reference mass equivalent, a conserved exterior compact charge, and a post-solution structural coordinate are *different physical observables.* Their connection follows from the fixed CPTG source–polarization–transport framework, the declared reduction of each physical state, conservation of the relevant charges, and matching of like physical quantities where the reductions overlap. That connection depends on comparing like physical quantities; it does not convert the arithmetic mass split into an independent law or treat N as a universal control of gravity.[2, 4, 7, 5]

The immediate consequence is methodological and physical. Observable gravitational behavior is characterized by a completed field conditioned on the material distribution, motion, support, and geometric state. A single material-mass figure does not supply all of that information. The subsequent galaxy and cluster sections accordingly evaluate registered reconstructions and their measured readouts, rather than inferring gravitational response from material inventory alone.

3 Resolved galaxies: a common field procedure across different dynamical states

The empirical starting point is not a catalog of invisible material. A resolved galaxy supplies positions, measured circular speeds, velocity errors, and gas and stellar gravitational-force templates. Together these measurements describe the observational state used in the registered galaxy reduction of CPTG. The stellar inventory is drawn from photometry; the gas inventory is reconstructed from the same signed SPARC gas-force template used in the gravitational source construction, but without consulting the completed CPTG field. The 153-primary-galaxy atlas applies that reduction to 3,168 SPARC radii, retaining the individual profiles and their actual observed uncertainties. The distinction made in Sec. 2 is indispensable here: the *published* geometric mass equivalent is an accounting readout of the completed field, not an independently inserted halo or a pointwise isolation of the polarization tensor.[4, 1]

3.1 A fixed-form radial method, evaluated on measured states

The parent equation (3) retains the Einstein and material sectors. The registered circular-field reduction takes a signed source template and locally observed support, evaluates its polarization carrier and transport factor, and solves a positive implicit field root. At each observed radius, define $Q(V) = V|V|$; the source and local structural acceleration are[4]

$$g_b(R_i) = \frac{Q(V_{\mathrm{gas,i}}) + Q(V_{\mathrm{disk,i}}) + Q(V_{\mathrm{bulge,i}})}{R_i}, \qquad a_\star(R_i) = \frac{V^2_{\mathrm{obs,i}}}{R_i}. \tag{18}$$

The unit-normalized disk and bulge templates in g_b are the inputs to the registered field solver. They are not the same as the calibrated material-force display or the independently calculated enclosed material inventory. In particular, signed gas contributions are preserved, rather than floored when a gas ring produces an outward contribution to radial force.

Writing $g_C = V_C^2/R$, the complete registered radial relation is[4, 3]

$$(1 + c_C\Sigma_C^2)g_C = g_b + A_C a_\star^{2/3} B_{\rm pol}\left[1 + \left(\frac{g_C}{a_\star}\right)^{123/50}\right]^{-11/50}. \tag{19}$$

The observed speeds and errors also determine the positive continuous support profile $g_{\rm sup}(R)$ via the registered uncertainty-weighted spline in $\ln R$. Primes denote analytic derivatives of this same support. The atlas explicitly defines the radial response operators[4]

$$\Sigma_C^2(R) = \frac{1}{3}\left[1 - \frac{d\ln g_{\rm sup}}{d\ln R}\right] \geq 0, \tag{20}$$

$$C_C(R) = g_{\rm sup}^2 + \kappa_C^2\left[(g'_{\rm sup})^2 + 2\left(\frac{g_{\rm sup}}{R}\right)^2\right], \tag{21}$$

$$B_{\rm pol}(R) = \frac{(R^2 + 2\kappa_C^2)g_{\rm sup}C_C^{-2/3} - \dfrac{d}{dR}\left[R^2\kappa_C^2 g'_{\rm sup}C_C^{-2/3}\right]}{R^2}. \tag{22}$$

These are the registered operators, not an alternative polarization law. Their contributions can be separated *within the same implicit equation* by defining

$$P_C(R) \equiv A_C a_\star^{2/3} B_{\rm pol}\left[1 + \left(\frac{g_C}{a_\star}\right)^{123/50}\right]^{-11/50}, \qquad t_C(R) \equiv c_C\Sigma_C^2(R), \qquad g_C = g_b + P_C - t_C g_C. \tag{23}$$

This identity exposes the radial balance: P_C enters as the polarization contribution to the solved acceleration, while $t_C g_C$ is its coupled transport contribution. Neither term is independently an enclosed material mass, and their difference must not be confused with the signed geometric mass remainder obtained by subtracting a separately reconstructed inventory. The measured source, support profile and geometric organization jointly determine the field.[4, 3]

The constrained continuous support supplies the admissible nonnegative Σ_C^2 without upper-clipping it to unity. The local $a_\star$, signed source, derivatives and positive implicit-root prescription remain exactly as registered; their reconstruction is conditioned on the measured dynamics. The radial-scale coefficient κ_C, polarization amplitude A_C, and transport coupling c_C remain *individual inherited reduced coefficients* from the registered v1.13.1 reconstructions. Their values are not universal constants. Neither these inherited coefficients nor the use of measured $V_{\rm obs}$ is to be concealed by calling the atlas a material-inventory-only forward calculation. No new coefficient optimization was made for the atlas mass split.[4]

For the material inventory reconstructed separately from the *CPTG field*, the atlas uses measured 3.6-μm disk photometry with $\Upsilon_{\star,\rm disk} = 0.5$, its declared bulge model with normalization 0.7, and a nonnegative gas-annulus inversion constrained by the *same signed* gas-force template used in the source construction. It then reports $M_{\rm material}(< R)$, which is not obtained by equating a flattened disk's radial force to the force of a spherical enclosed mass. The field output is converted, *for accounting*, to $M_{\rm grav}^{\rm eq} = RV_C^2/G$, and the difference $M_{\rm geom}^{\rm eq} = M_{\rm grav}^{\rm eq} - M_{\rm material}$ follows the conventions of Eqs. (10)–(13).[4]

3.2 Structural Mode of a resolved galactic field

For resolved galaxies, the continuous structural carrier and positive weight are[5]

$$C_{N,\rm gal}(r) = g_C^2(r) + \ell_{\rm s}^2\left(\frac{dg_C}{dr}\right)^2, \qquad w_{N,\rm gal}(r) = \left|\frac{dC_{N,\rm gal}}{dr}\right|, \quad \ell_{\rm s} = 1\,{\rm kpc}. \tag{24}$$

Its continuum outer-field readout is

$$\lambda_{N,\rm gal} = \frac{\displaystyle\int_{R/5}^{R} r\, w_{N,\rm gal}(r)\, dr}{\displaystyle\int_{R/5}^{R} w_{N,\rm gal}(r)\, dr}, \qquad N_{\rm gal} = \frac{R}{\lambda_{N,\rm gal}}. \tag{25}$$

The registered *sampled* SPARC estimator is distinct from the continuum integral: it computes gradients on the actual nonuniform grid, applies its specified five-point smoothing where appropriate, and forms an unweighted sum over retained outer sample points, $\lambda_{\rm num} = \sum_i r_i w_i / \sum_i w_i$. There is no inserted Δr_i quadrature factor. The published 153-galaxy mode values are those of this sampled estimator; their mean is 2.237438 and median 2.150980.[5]

3.3 The published result across 153 primary galaxies

The primary sample retains SPARC quality-selected galaxies after removing catalog objects flagged $Q = 3$ or with inclination below 30°; there is no further radius-level selection in the atlas. All published geometric-equivalent percentages below refer to each object's *last measured radius*, which varies from galaxy to galaxy. The statistics are therefore neither a common-aperture density measurement nor a cosmic matter fraction.[4]

Published outer-radius atlas quantity	**Registered result**
Primary galaxies; observed radii	153; 3,168
Unweighted mean geometric-equivalent share	77.96%
Median; interquartile interval	80.33%; 73.01–86.46%
Mass-equivalent-weighted geometric share	79.03%
Sum of total field equivalents	$28.09368 \times 10^{12}\ M_\odot$
Sum of material inventories	$5.89138 \times 10^{12}\ M_\odot$
Sum of geometric mass equivalents	$22.20230 \times 10^{12}\ M_\odot$

Table 2: Published Register A, retained without re-estimation or renormalization. The sums aggregate 153 *different* sampled outer apertures; the 79.03% share uses the sum of gravitational field equivalents as denominator. Values are from the authoritative 153-SPARC publication.

The totals close arithmetically: $28.09368 = 5.89138 + 22.20230$, all in $10^{12} M_\odot$. The last-row ratio gives the published mass-weighted fraction after rounding. This numerical closure is a verification of the declared mass ledger. It does not independently establish the gravitational field equation or demonstrate that every rotation curve is reconstructed equally well.[4]

3.4 Signed values, observational radii, and what the percentages mean

The subtraction in Eq. (11) is signed. A negative remainder states that, under the adopted material inventory and spherical-reference convention, the material aperture mass exceeds the completed-field gravitational equivalent at that radius. It does not establish negative material density or a separate negative-mass substance. The atlas preserves the signed number; where $M^{\rm eq}_{\rm geom} < 0$, its displayed real-valued geometric speed, $\sqrt{GM^{\rm eq}_{\rm geom}/R}$, is *omitted*, not forced to zero. The signed catalog counts and observational-systematic consequences are reported below alongside the full galaxy results.[4]

The catalog-level percentages in Table 2 retain their aperture meaning. The 153 primary galaxies contribute one outermost measured radius $R_{{\rm max},g}$ each; their mass-equivalent-weighted outer fraction is[4]

$$\frac{\sum_{g=1}^{153} M^{\rm eq}_{{\rm geom},g}(R_{{\rm max},g})}{\sum_{g=1}^{153} M^{\rm eq}_{{\rm grav},g}(R_{{\rm max},g})} = \frac{22.20230 \times 10^{12} M_\odot}{28.09368 \times 10^{12} M_\odot} \simeq 79.03\%. \tag{26}$$

These sums combine *different observed radii and different physical volumes*. In view of Eq. (15), the 79.03% result must not be presented as a direct measurement of the fraction of total gravitational force generated specifically by polarization–transport. Its reported definition is the aggregate of the atlas's declared mass-equivalent accounting remainders. They are neither a cosmic matter-density fraction nor the mass of one gravitationally bound aggregate. The atlas further reports six catalog entries with velocity-χ^2 replay differences above 0.01, fourteen material/proxy flags, and no full joint distance, inclination, and material-profile systematic propagation. Their presence must remain visible when the reconstruction statistics are interpreted.[4]

3.5 Different galaxies, different outer-field mass equivalents

The atlas's own examples show that the method does not impose one geometric percentage on every object. Table 3 reproduces four *published* outer-radius results; each row uses the same radius for its three mass columns, but the radii *differ between galaxies.*[4]

Galaxy	$R_{\max}$ (kpc)	$M^{\rm eq}_{\rm grav}$	$M_{\rm material}$	$M^{\rm eq}_{\rm geom}$	$f_{\rm geom}$
DDO 154	5.92	2.8590	0.2966	2.5624	89.63%
UGC 00731	10.91	14.7322	2.2621	12.4701	84.64%
NGC 7331	36.31	451.9758	146.6977	305.2780	67.54%
NGC 2955	35.43	451.5476	207.7179	243.8297	54.00%

Table 3: Four original atlas examples, copied at the precision of the publication. All mass columns are in $10^9 M_\odot$. Rounding means that printed components need not sum to the final printed digit. These are Register A values, not separately measured net geometric accelerations.

For DDO 154 the last radius is 5.92 kpc; the original SPARC speed is 45.5 ± 1.3 km s^{-1} and the registered CPTG solution gives $V_C = 45.575126$ km s^{-1}. The matched-radius account yields $M_{\rm material} = 0.296579 \times 10^9 M_\odot$, $M^{\rm eq}_{\rm grav} = 2.859015 \times 10^9 M_\odot$, and $M^{\rm eq}_{\rm geom} = 2.562436 \times 10^9 M_\odot$, or 89.6265%. The observed speed is an explicit *input* to its local structural acceleration and to the support interpolation, so the close outer-speed result is a reconstruction check, not an independent prediction withheld from the solver. NGC 2955, with a different distribution and dynamical state, gives 54.00% under the same accounting rules.[4]

3.6 Atlas results and observational systematics

The original atlas retains unfavorable entries. In 49 galaxies, a total of 196 sampled radii have *negative signed arithmetic remainders* $M^{\rm eq}_{\rm geom}$; these are not discarded or changed to zero. A non-real square-root speed equivalent is simply not plotted at those positions. Fourteen galaxies have reconstructed material inventories exceeding a catalog-wide photometric/H I proxy by more than 30%, and six galaxies have replay χ^2 differences larger than 0.01 relative to the registered baseline. None of these facts is silently removed from the population result.[4]

Reconstruction residuals are also nonuniform: 33 of the 153 galaxies have $\chi^2/N_{\rm radii} > 2$ and 14 exceed 10. The ratio is not a formal reduced χ^2 because an effective fitted degrees-of-freedom count and full covariance have not been established. A successful arithmetic decomposition is not evidence of an accurate kinematic reconstruction in a galaxy with a large velocity residual. Nor have the publication's distance, inclination, and material-profile systematics been jointly propagated through the full population.[4]

The important physical result is more specific than a universal numerical fraction. One registered source–support–polarization–transport procedure, with the individual fitted coefficients disclosed above, maps observed dynamical states to completed reconstructed fields. Material inventories reconstructed without using those completed fields then permit a consistent geometric-mass-*equivalent* accounting, although the gas inventory and field source share the original measured gas-force template. This connects the field construction to diverse resolved dynamical systems. The physical interpretation does not require a new material halo in the CPTG field equation, while the degree of independent discrimination from other descriptions depends on the information used in the reconstruction.[4, 2]

3.7 Two distinct mass ledgers in galaxy reconstruction

The galaxy reconstruction in Section 3 and its radial-force continuation retains *two explicitly separate registers*; the additional force comparison does not replace the published atlas accounting. Table 4 distinguishes each quantity and its comparison basis. Both use the same completed CPTG field and the same physical radius when evaluated for an individual galaxy, but compare that field with different representations of ordinary material. They are not two independent confirmations of the gravitational field, and neither is an independently inventoried dark-material mass.

Register	Quantity and physical comparison	Basis of comparison
A: Atlas mass equivalent	$M^{\rm eq}_{\rm geom} = M^{\rm eq}_{\rm grav} - M_{\rm material}(< R)$; completed-field spherical equivalent versus independently reconstructed material aperture inventory.	Published atlas mass-inventory comparison; signed entries, outer radii and 79.03% weighted share.
B: Field–force equivalent	$\Delta M^{\rm eq}_{\rm field\|force} = R^2(g_C - g_{\rm mat,force})/G$; completed field versus calibrated signed material radial force.	Separate 153-galaxy radial-force diagnostic using the registered field and signed source-force inputs; weighted outer share 76.8775%.

Table 4: Two-register comparison for resolved galaxies. Register A retains the published atlas accounting; Register B is a subsequently calculated radial-force diagnostic from verified source and field arrays. Neither register is inferred from the other.

The identity in Eq. (15) relates the registers through a third, geometry-and-inventory conversion remainder, $\Delta M^{\rm eq}_{\rm force|inventory}$. Register B does *not* inherit the atlas's 77.96% mean, 80.33% median, or 79.03% mass-equivalent-weighted outer share: those published values belong exclusively to Register A. The subsequently verified radial calculation reports its own values, signs, uncertainties and provenance in the radial-force continuation of Section 3. The first-register arithmetic and catalog quality flags remain untouched. Both registers remain observationally conditioned on the *same* completed CPTG gravitational reconstruction.[4]

3.8 A separate field–versus–material-force diagnostic

Register B is evaluated separately from the original signed SPARC gas, disk, and bulge velocity templates at all 3,168 published radial locations. It shares the *same completed CPTG field* with Register A; it differs only in the reference subtracted from that field. It is therefore a distinct accounting diagnostic, *not* an independent observational replication or a substitute for the published Register A mass-inventory statistics. The signed calibrated material-force reference is

$$g_{\rm mat,force}(R_i) = \frac{V_{\rm gas,i}|V_{\rm gas,i}| + 0.5V^2_{\rm disk,i} + 0.7V^2_{\rm bulge,i}}{R_i}, \qquad \Delta M_{\rm field-force}(R_i) = \frac{R_i^2}{G}\left[g_C(R_i) - g_{\rm mat,force}(R_i)\right]. \tag{27}$$

The force reference applies the adopted stellar mass-to-light coefficients to SPARC's *signed* source-force templates; it is not the unit-normalized source entering Eq. (19). Nor is it a separately counted enclosed physical mass: the disk-force equivalent adopts a spherical reference while the independently reconstructed inventory integrates the declared flattened source profile. The gas-force template also enters the regularized inventory inversion; consequently the difference between ledgers includes the effects of the stated inventory-model conventions, not just source flattening. The sample-wide and Milky Way calculations use the same two-ledger identity as Eq. (15), with all quantities evaluated at matched radii. Register B is a separate radial-force diagnostic evaluated from the previously registered fields at 3,168 SPARC radii and 45 Milky Way radii (3,213 radial evaluations in total); it is not an independently observed material component.[4, 1]

Outer-radius population statistic	Result
Register A: published mean / median	77.96% / 80.33%
Register A: published weighted share	79.03%
Register B: mean / median	73.93% / 77.02%
Register B: weighted share	76.88%
Register A minus B, weighted shares	+2.15 percentage points
Negative radial Register A / B entries	196 / 108
Galaxies with negative radial A / B entries	49 / 22

Table 5: Two distinct accounting diagnostics of the *same* 153-galaxy CPTG reconstructed fields, assessed at 3,168 radial locations. Register A is reproduced from the published 153-galaxy analysis; Register B is newly evaluated using the calibrated signed SPARC material-force template and the *registered* CPTG field, with no refitting. The weighted fractions are ratios of sums of equivalents at each galaxy's own last observed radius. They are not independent fits, cosmic matter fractions, universal geometric response coefficients, or observations at one common physical aperture.

At the last sampled radius of each galaxy the newly evaluated Register B has a mean of 73.9275%, a median of 77.0243%, and a mass-equivalent-weighted share of 76.8775%. It contains 108 negative signed values at 22 galaxies over the full 3,168-radius sample, with no negative endpoint values. The original atlas Register A retains its 196 negative radial values at 49 galaxies. No negative remainder was clipped. The difference between the weighted shares is 2.1520 percentage points. Algebraically, $M^{\rm eq}_{\rm geom,A} - \Delta M_{\rm field-force} = M^{\rm eq}_{\rm mat,force} - M_{\rm material}$ at each radius; the weighted gap therefore belongs to the *force-equivalent versus modeled-inventory convention*, not to an additional CPTG force fitted to the observations. Its sign can vary between galaxies.

Galaxy	$R_{\max}$ (kpc)	Register A share	Register B share	$\Delta M_{\mathrm{field-force}}$
DDO 154	5.92	89.63%	87.28%	2.4954
UGC 00731	10.91	84.64%	76.58%	11.2814
NGC 7331	36.31	67.54%	65.70%	296.9325
NGC 2955	35.43	54.00%	49.51%	223.5616
UGC 01281	4.99	61.56%	80.03%	2.7190

Table 6: Matched-radius outer readouts for five catalog members. The final column is in $10^9 M_\odot$. Register B uses the signed, physically scaled SPARC radial material-force templates, not the nonnegative inversion used to reconstruct the enclosed gas mass inventory. UGC 01281 shows explicitly that the force-equivalent-minus-inventory remainder can be negative; the ledgers must not be merged into one statistic.

This separate force diagnostic sharpens the interpretation of the mass discrepancy, but it is not an isolated numerical measurement of the covariant polarization or transport tensor: the CPTG field is still obtained from the registered observation-conditioned source–support procedure, and the signed SPARC material-force template has its own photometric, inclination, distance and gas-model assumptions. The newly evaluated quantities are deterministic readouts from *unchanged* recorded radial data; they are neither a new field solve nor a fit to an independent observation sample. No joint uncertainty interval is claimed for the new population share.

3.9 The Milky Way: corrected material geometry and two force ledgers

The Milky Way offers a different kind of source reconstruction. Rather than treating integrated disk luminosity as a spherical force profile, the CPTG source study takes the thin and thick stellar disks, atomic and molecular gas layers, and oblate bulge from McMillan's specified three-dimensional material model. The conventional halo component of that model is not included as a CPTG material source.[6, 38] The midplane bulge force uses the corrected homeoidal integral,

$$g_{R,\mathrm{bulge}}(R,0) = \frac{4\pi G q}{R}\int_0^R \frac{\rho_{\mathrm{bulge}}(s)s^2\,ds}{\sqrt{R^2-(1-q^2)s^2}}, \qquad q = 0.5. \tag{28}$$

The independent axisymmetric finite-volume Poisson check agrees with the combined material midplane force across the 45 published rotation-curve bins to approximately 0.0036% at its finest resolution. This is a *source-field* check, performed separately from the CPTG field reconstruction. The observed inputs are the 45-bin published Gaia DR3+ rotation-curve compilation, spanning 5.25 to 27.25 kpc and assembled using previously reported Galactic kinematic studies; these are radial bins rather than 45 newly measured individual stars.[6, 39, 40, 41]

Using the registered CPTG solution without readjusting its response coefficients gives an in-sample diagonal-error statistic $\chi^2_{\mathrm{diag}} = 7.347858$ and an RMS speed residual of $1.719\,\mathrm{km\,s^{-1}}$. At the outer measured radius the same reconstructed circular field has $V_C = 181.1575\,\mathrm{km\,s^{-1}}$ and gravitational equivalent $2.079304 \times 10^{11} M_\odot$. Its physical material inventory is $6.499360 \times 10^{10} M_\odot$.[6]

Milky Way, $R = 27.25\,\mathrm{kpc}$	**Value**
CPTG total gravitational mass equivalent	$2.079304 \times 10^{11} M_\odot$
Physical material inventory	$6.499360 \times 10^{10} M_\odot$
Register A geometric remainder	$1.429368 \times 10^{11} M_\odot$
Register A geometric share	68.7426%
Register B field–minus–material-force equivalent	$1.369389 \times 10^{11} M_\odot$
Register B geometric share	65.8580%
Force–inventory conversion remainder	$5.997944 \times 10^{9} M_\odot$
Structural Mode (post-solution)	$N_{\mathrm{MW}} = 2.826813$

Table 7: Milky Way quantities on the same 27.25 kpc radial boundary. Register A and Structural Mode follow the Milky Way calculation; Register B and the force–inventory remainder are separately evaluated arithmetic diagnostics from the stated 45-radius source and field conventions. The material-force reference is independently modeled and finite-height, not a spherical reconstruction of its material inventory.

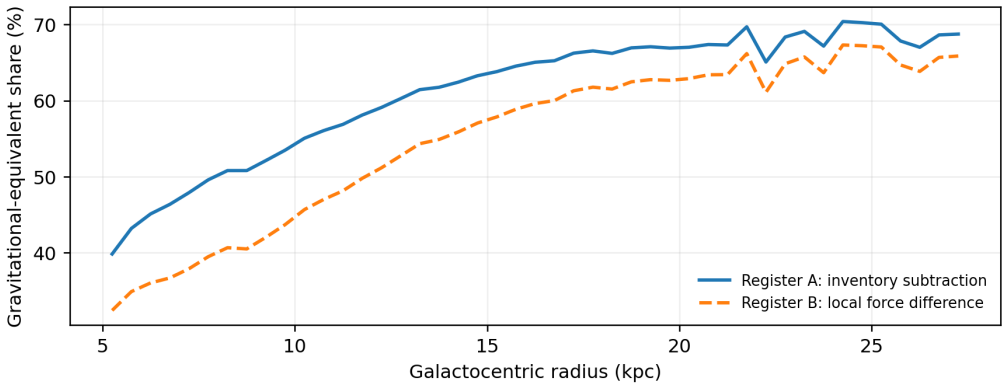


Figure 1: Radial two-ledger gravitational-equivalent fractions in the Milky Way, reconstructed at the 45 published radii. Both curves use the same completed CPTG circular field; their difference arises entirely from whether one subtracts the independently integrated material inventory (Register A) or the independently computed finite-height material *force* converted to a spherical-reference equivalent (Register B).

The 2.8846 percentage-point difference between the two Milky Way readings is a quantified difference between the finite-height material-force equivalent and the adopted enclosed material inventory, not an adjustment of the CPTG polarization or transport law. It depends on the source geometry *and* on the registered inventory and force conventions. In the published independent Gaia B2 comparison, a conventional Einasto halo encloses $1.436575 \times 10^{11} M_\odot$ at the same radius, whereas the CPTG *Register A* geometric equivalent is $1.429368 \times 10^{11} M_\odot$. The close arithmetic values belong to the stated, distinct material models and the same radial aperture; they do not demonstrate identical radial force decompositions or remove the observational uncertainties of either model.[6]

Both investigations distinguish enclosed material *inventory* from signed material *force* in a nonspherical source. A spherical-reference conversion gives common mass units, not identical physical integrals. Both ledgers use the same completed CPTG field; neither uniquely isolates a covariant polarization or transport contribution. Structural Mode remains a post-solution readout, never a parameter used to make a ledger close.

4 Pressure-supported systems: a change of state, not a change of law

A rotating disk is read primarily through orbital speed; a hot intracluster gas atmosphere is read through thermodynamic pressure gradients; and a dispersion-dominated stellar galaxy is read through projected velocity moments. These are different observations of different dynamical states, not interchangeable ways of assigning unseen material. In CPTG, the material distribution and its motion remain part of the measured state, the Einstein and material terms remain continuously present, and the registered polarization–transport response is evaluated using the appropriate support geometry. This section therefore tests the reach of the common gravitational architecture without substituting the circular-speed reduction for hydrostatic or stellar-stress equations.[3, 2]

4.1 Support reconstruction and the material source

For a spherical gas atmosphere satisfying the declared hydrostatic approximation, the positive inward support acceleration can be written

$$g_{\rm sup}(r) = -\frac{1}{\rho_{\rm gas}(r)}\frac{dP(r)}{dr}, \qquad a_\star(r) = g_{\rm sup}(r). \tag{29}$$

The measured $P(r)$ and $\rho_{\rm gas}(r)$ specify the gas-support state. Identifying the support with the inward gravitational acceleration assumes the adopted spherical hydrostatic balance; unmodeled nonthermal pressure or departures from equilibrium can change that identification. Thus $a_\star$ is an *observation-conditioned support coordinate*, not a new material source or a universal constant. The separately constructed material inventory is

$$M_b(<r) = M_{\rm gas}(<r) + M_\star(<r), \qquad g_b(r) = \frac{GM_b(<r)}{r^2}. \tag{30}$$

The common reduced-field reconstruction in the seven-cluster program is built from these distinct support and material

inputs. For a spherical support-equivalent mass $M_{\rm sup} = r^2 g_{\rm sup}/G$, the corresponding shape carrier is[3]

$$\Sigma_C^2(r) = \frac{1}{3}\left(1 - \frac{d\ln g_{\rm sup}}{d\ln r}\right). \tag{31}$$

This relation expresses the radial shape carried by the measured support field in the registered spherical reduction. It neither turns gas density into additional gravitational matter nor independently determines the gravitational response without the reconstructed support state. Spherical balance, adopted thermodynamic profiles, the retained radial domain, and possible disequilibrium remain explicit assumptions; recovered reduction coefficients are object-conditioned coordinates, not independently determined universal constants.[3]

For a collisionless, spherical isotropic or anisotropic stellar system, a Jeans reconstruction instead relates tracer density ν, radial velocity dispersion σ_r, and anisotropy β to the inward gravitational field,

$$g(r) = -\frac{1}{\nu(r)}\frac{d[\nu(r)\sigma_r^2(r)]}{dr} - \frac{2\beta(r)\sigma_r^2(r)}{r}. \tag{32}$$

For an axisymmetric stellar source, the coupled Jeans tensor equations and viewing geometry replace the spherical shorthand; the observed second moment is obtained only after projection through the appropriate aperture. This shift from orbital speed to pressure or velocity moments is physically important: it changes *how the same gravitating field is read*, not which material is permitted to source the Einstein term. A stellar dispersion is measured dynamical-state information, not an additional material source. Nor is using the same dispersion to establish the support coordinate and check the resulting aperture value an independent held-out test.[9, 10]

4.2 Seven clusters: X-COP and ACCEPT remain distinct

The registered cluster sample comprises A2029, A1795, A2142, A2319, A644, A85, and ZW1215. The original X-COP analysis uses 294 radial support measurements over the individually retained intervals; A2029 uses 60–670 kpc, while the other six use their declared inner $0.47R_{500}$ domains. Its reported aggregate statistic is[3]

$$\chi^2_{\text{X-COP}} = 266.114, \qquad \nu = 273, \qquad \chi^2_{\text{X-COP}}/\nu = 0.975. \tag{33}$$

The corrected ACCEPT reconstruction changes the pressure/density data stream, retains the *same baryonic source grid*, and propagates its adopted 400-draw pressure and density uncertainty model into a correlated support covariance. It uses 238 radial measurements and reports[3]

$$\chi^2_{\text{ACCEPT,cov}} = 102.029497, \qquad \nu = 217, \qquad \chi^2_{\text{ACCEPT,cov}}/\nu = 0.470182. \tag{34}$$

The two aggregates describe *different* radial records and uncertainty models; their ratios are not a shared likelihood or a measure of relative model superiority. ACCEPT provides a separate *thermodynamic-support* reconstruction but inherits the X-COP material-source grid. Its small covariance-weighted ratio must be interpreted with the stated 400-draw covariance and should not be read as a claim of extraordinarily precise independent material accounting.[3]

Cluster	$n_{\rm X}$	X-COP median $\lvert\Delta\rvert/\sigma$	$n_{\rm A}$	ACCEPT median $\lvert\Delta\rvert/\sigma$
A2029	52	0.118	46	1.0412
A1795	41	1.540	33	0.6316
A2142	40	0.663	29	0.6504
A2319	40	0.098	33	2.3240
A644	40	0.118	37	0.5348
A85	41	1.076	22	0.0709
ZW1215	40	0.033	38	0.1802

Table 8: Seven-cluster support reconstructions as reported in the registered CPTG gravitational-law paper. $n_{\rm X}$ and $n_{\rm A}$ are retained radial sample counts; medians refer to each paper's stated diagonal uncertainty normalization, not a joint likelihood evaluated on matched radii with a common covariance. The ACCEPT baryonic source grid is inherited from X-COP.

The difficult objects remain visible. X-COP A1795 and A85 have median absolute pulls above 1σ and reported reduced χ^2 values of 3.649 and 2.031, respectively; their maximum pulls reach 4.370σ and 3.527σ. A2319 has an X-COP median of only

0.098σ but an ACCEPT median of 2.3240σ (maximum 3.1030σ). Its disturbed, merging state puts the adopted spherical equilibrium description under stress. Several recovered X-COP coefficients also lie at numerical branch boundaries. These qualifications are part of the published reconstruction, not grounds to discard less favorable measurements.[3]

4.3 Cluster Structural Mode: a measurement of each completed field

For the solved spherical cluster field $g_C(r)$, the registered post-solution invariant and weight are[3]

$$C_{N,C}(r) = 3g_C^2(r) + \left[r\frac{dg_C}{dr}\right]^2, \qquad w_{N,C}(r) = \left|\frac{dC_{N,C}}{dr}\right|, \tag{35}$$

with the spherical-shell organization length and Structural Mode

$$\lambda_{N,C} = \frac{\displaystyle\int_{R/5}^{R} r^3 w_{N,C}(r)\,dr}{\displaystyle\int_{R/5}^{R} r^2 w_{N,C}(r)\,dr}, \qquad N_C = \frac{R}{\lambda_{N,C}}. \tag{36}$$

As the registered Structural Mode source specifies, unlike the disk readout this operator uses the cluster's own spherical shell measure, sampled radial domain, and field gradient. N is evaluated *after* the support-conditioned field has been solved; it neither supplies gravitational support nor chooses a response branch.

Cluster	$N_{\text{X-COP}}$	N_{ACCEPT}
A2029	1.521388	1.397476
A1795	1.566115	1.507837
A2142	1.328084	1.760504
A2319	1.350667	1.400538
A644	1.400836	1.120615
A85	1.533717	1.428954
ZW1215	1.358032	1.304734

Table 9: Object-specific post-solution Structural Mode readouts from the original X-COP and the separately reconstructed ACCEPT fields. They must not be averaged or pooled; domain and support changes legitimately change the field organization being measured.

A2142 and A644 show especially large differences between the two reconstructions. Those shifts reflect the fields and domains actually reconstructed, and must not be eliminated by averaging into an artificial "cluster mode." What is held common is the post-solution measurement rule. No N value is used to adjust the force law or to choose a preferred X-COP or ACCEPT solution.[3, 5]

4.4 Dragonfly 44: axisymmetric stellar support

Dragonfly 44 supplies a complementary test because its observed stellar system is markedly nonrotating, flattened, and resolved into nine Keck Cosmic Web Imager (KCWI) apertures. Its source model uses an oblate deprojection of the measured Sérsic light profile, evaluates the material potential in that geometry, and constructs a conservative axisymmetric CPTG extension. One *object-level* structural acceleration is conditioned on the reported integrated half-light second moment $\sigma_{1/2} = 33 \pm 3\,\mathrm{km\,s^{-1}}$; no separate acceleration is assigned to each resolved aperture.[9]

For the primary edge-on neutral branch, the paper reports

$$a_\star = 5.294 \times 10^{-10}\,\mathrm{m\,s^{-2}}, \quad \chi^2_{\text{diag,CPTG}} = 9.21, \quad \chi^2_{\text{diag,stars}} = 539.66, \quad N_{\text{DF44}} = 1.655926168. \tag{37}$$

All nine reported second-moment residuals in the primary geometry lie within 1.47σ. The stellar-only aperture dispersion is $6.14\,\mathrm{km\,s^{-1}}$ in the declared comparison, substantially below the measured $33\,\mathrm{km\,s^{-1}}$ half-light value. The resolved second moments are conditional on the common half-light support normalization, and the quoted diagonal statistic is not a covariance-complete model-comparison likelihood.[9]

The minimal two-integral fourth-moment continuation does not reproduce observed $h_4 = 0.13 \pm 0.05$. With h_4 *withheld*, two three-integral orbit libraries in the fixed CPTG potential instead yield -0.00416 and -0.02164. Nonnegative orbit weights recover positive h_4 only when it is supplied as an additional phase-space constraint. This establishes conditional compatibility of the fixed potential, not independent recovery of the higher moment or success of the minimal closure. Structural Mode remains a post-field measurement.[9]

4.5 NGC 1052-DF2 and DF4: low-response diffuse galaxies

NGC 1052-DF2 provides the published low-response counterpart. At the adopted JWST distance 17.6 ± 0.6 Mpc, the source model has $M_\star = 1.70368 \times 10^8 M_\odot$, a major-axis effective radius $R_e = 1.92839$ kpc, and Sérsic index $n = 0.6$. The updated MUSE integrated stellar dispersion $9.2^{+3.8}_{-4.5}\,\mathrm{km\,s^{-1}}$ establishes the declared support coordinate; the isotropic stellar-only and CPTG aperture dispersions are 7.22767 and $9.19500\,\mathrm{km\,s^{-1}}$, respectively. At R_e, $g_{\mathrm{CPTG}}/g_\star = 1.55102$, and the completed field gives $N_{\mathrm{DF2}} = 1.56680$.[10]

This agreement is *shared-input closure*: MUSE also fixes $a_\star$. KCWI's $6.3^{+3.1}_{-3.7}\,\mathrm{km\,s^{-1}}$ measurement has a different footprint, not reproduced exactly by the spherical projection. The stellar-only $7.22767\,\mathrm{km\,s^{-1}}$ result also lies within the broad observed errors; the comparison is conditionally consistent, not a unique detection of geometric enhancement. The local R_e acceleration ratio is not an integrated mass fraction.[10]

A subsequent, separate *exploratory* DF4 spherical isotropic calculation applies the registered DF2 reduction to DF4's *own* structural inputs: $D = 20$ Mpc, $M_\star = 1.5 \times 10^8 M_\odot$, $R_e = 1.6$ kpc, $n = 0.79$, and an updated KCWI support value $7.8^{+2.3}_{-2.0}\,\mathrm{km\,s^{-1}}$. Its calculated circular-aperture dispersion is $8.81090\,\mathrm{km\,s^{-1}}$, with $g_{\mathrm{CPTG}}/g_\star = 1.37364$ at R_e and $N_{\mathrm{DF4}} = 1.54526$. Under the separately documented alternative inner-light profile, N is approximately 1.63816; the two source profiles are not mixed.[11, 12, 13]

This DF4 extension remains an *exploratory spherical calculation*, not an independently established test of the full observational geometry. The measured dispersion both establishes the support scale and anchors its aperture comparison. The model's circular aperture does not reproduce the exact KCWI footprint; tidal light and the external group field are also unsolved. The alternate photometric profile is a separate sensitivity case. Radius-specific mass equivalents are not galaxy-integrated fractions.[11]

4.6 What the pressure-supported tests add

These tests extend CPTG reconstruction from rotating disks to collisional gas pressure in clusters and collisionless stellar velocity stresses in diffuse galaxies. Their different observational interfaces retain the same proposed Einstein–polarization–transport architecture but demand appropriate support equations, source geometry, covariance, and aperture projections. The residuals reflect reconstructions under these conditions; incompatible χ^2 statistics cannot be pooled into a cross-regime significance.

Structural Mode provides a separate post-solution fingerprint, not a replacement for support measurements or a parameter adjusting the gravitational field. The shared mathematical principles must be established while retaining each object's distinct dynamical state.[3, 9, 10, 2]

5 One Native Cosmological History and Its Observational Coordinates

Cosmological observations test a different aspect of the same framework. Where a bound object is reconstructed from its measured material and dynamical state, the cosmological papers begin with one native geometric history and translate it into specified observational coordinates. An acoustic distance, a luminosity distance, a native horizon radius, and a CMB comparison map are therefore neither four independently selected expansion histories nor four measurements of the same observable. Each has a declared input and output. The common source–polarization–transport architecture is retained while the observation coordinate changes. In particular, the native history produces late-time cosmic acceleration from geometric branch evolution without a separate cosmological-constant or dark-energy fluid.[2, 14, 15]

The observational mappings have a defined sequence. The Hubble bridge first fixes the native, acoustic, and luminosity endpoint relations. The DESI calculation carries the already registered acoustic branch into a redshift-dependent distance-

ratio vector using an observed CMB acoustic boundary. The horizon study evaluates native geometric containment at that boundary's epoch. The map study independently translates a fixed geometric-pi comparison row into an angular-power envelope and, using observed harmonic phases, into a temperature comparison map. The common theory-side construction is the connection; the observation-derived inputs of these calculations must not be conflated.

5.1 Native geometric phase and projected clocks

The common geometric stage is denoted $N_\pi = \ln a$, to distinguish it from the *post-solution* Structural Mode N of bound gravitational objects. The native branch and the acoustic and luminosity comparison branches share this stage label while accumulating their own declared branch-clock coordinates:

$$H_b\, dt_b = dN_\pi, \qquad b \in \{\pi, \mathrm{ac}, L\}. \tag{38}$$

A common N_π identifies a corresponding geometric stage; it does not assert identical accumulated clock readings in all three observational coordinates. The Hubble-bridge construction uses a fixed native normalization and its exact acoustic map. With $p_{\mathrm{ac}} = 3 - \pi/100$ and $G_T = p_{\mathrm{ac}}/\pi$, the registered nonconstant contribution transforms as[15, 14]

$$X_{\mathrm{ac}}(a) = T_\pi^{1-G_T}\,[X_\pi(a)]^{G_T} = T_\pi a^{-p_{\mathrm{ac}}}, \qquad X_\pi(a) = T_\pi a^{-\pi}. \tag{39}$$

This acoustic projector preserves the normalization of the nonconstant native contribution at $a = 1$. The separately registered affine luminosity projector uses $G_L^{\mathrm{aff}} = 10/9$ and $\Gamma_L^{\mathrm{aff}} = \sqrt{10/9}$ to change the comparison-clock rate without altering the native history. In particular, the luminosity endpoint is not obtained by adjusting the acoustic endpoint to a local observational value. The meaningful connection between the subsequent tests is their declared dependency on this fixed branch architecture, not a numerical resemblance among fitted rates.

5.2 Cosmic acceleration from native curvature transport, without a dark-energy fluid

The late-time acceleration of the Universe is encoded in the native curvature–transport history rather than added as a separate vacuum-energy source. The universe-scale reduction uses the fixed curvature exponent $p_C = \pi$ and its registered complementary geometric weights[2, 14]

$$p_C = \pi, \qquad B_\pi = 1 - \frac{1}{p_C}, \qquad T_\pi = \frac{1}{p_C}, \qquad B_\pi + T_\pi = 1. \tag{40}$$

The parent cosmological construction specifies this partition as part of its registered native branch; it is not a separate observational fit or an independently demonstrated full-action derivation of the weights. Neither term represents a material-density fraction or a vacuum-energy fluid. The native history then follows the declared branch law[2, 14]

$$E_\pi(a)^2 = B_\pi + T_\pi a^{-p_C} = \left(1 - \frac{1}{\pi}\right) + \frac{a^{-\pi}}{\pi}, \qquad H_\pi(a) = H_0^{(\pi)} E_\pi(a). \tag{41}$$

The transport-dependent term dominates early; its diminishing share leaves the nonzero late-time geometric background $H_\infty = H_0^{(\pi)}\sqrt{B_\pi}$. The registered background curvature–transport radius changes correspondingly as $R_C(a) = R_{C,0}/E_\pi(a)$. Differentiating this *specified history* gives the instantaneous transport share and native deceleration parameter:

$$f_\pi(a) = \frac{T_\pi a^{-\pi}}{E_\pi(a)^2}, \qquad q_\pi(a) \equiv -1 - \frac{d\ln H_\pi}{d\ln a} = -1 + \frac{\pi}{2} f_\pi(a). \tag{42}$$

At the present native stage $q_{\pi,0} = -1/2$. Setting $q_\pi = 0$ in Eq. (42) equates the decelerating transport share to the persistent geometric contribution. The transition follows algebraically:

$$a_{\mathrm{tr}}^{-\pi} = \frac{2(\pi-1)}{\pi-2}, \qquad z_{\mathrm{tr}} = \left[\frac{2(\pi-1)}{\pi-2}\right]^{1/\pi} - 1 = 0.5233204280, \qquad a_{\mathrm{tr}} = 0.6564607036. \tag{43}$$

Thus the registered native curvature–transport history evolves continuously from deceleration to acceleration without postulating an additional dark-energy fluid. These acceleration values follow by differentiation once the geometric branch is specified, independently of the later acoustic and luminosity observational projections.

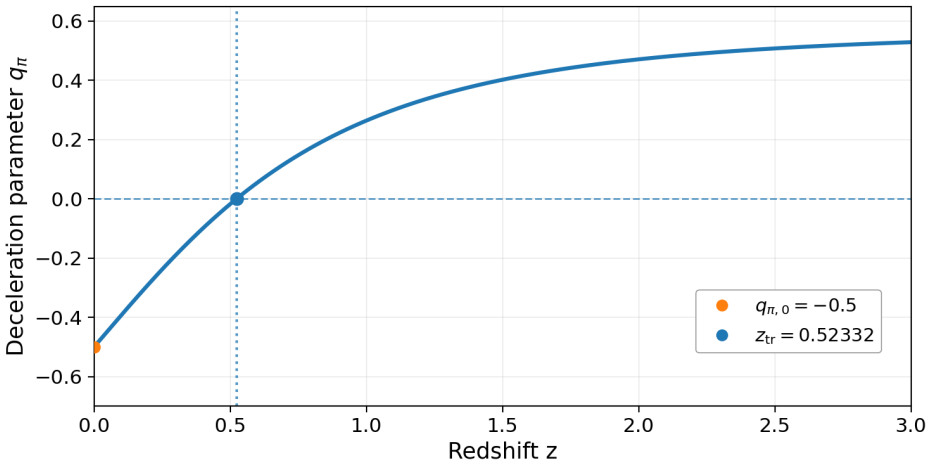


Figure 2: Native geometric deceleration $q_\pi(z)$ evaluated directly from the published fixed branch. The zero crossing at $z = 0.523320428$ separates the earlier decelerating history from the accelerating late-time regime; no observational fit enters this curve.

Ordinary material and radiation sources remain part of the CPTG gravitational framework; the extra cosmic response is geometric. This removes the need to tune a separate dark-energy density to generate acceleration within the registered native construction.[2, 14]

5.3 The Hubble bridge: two observational endpoints

The Hubble-bridge paper reports the following native and projected present-epoch quantities, in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$:[15]

$$H_0^{(\pi)} = 69.4162507897, \qquad H_{0,\mathrm{ac}} = 67.4777967351, \qquad H_{0,L}^{\mathrm{aff}} = 73.1711530416. \tag{44}$$

The acoustic and luminosity endpoints differ by 8.43737730% relative to the acoustic value. The source paper compares them, respectively, with Planck 2018's 67.4 ± 0.5[43] and the published four-anchor SH0ES value 73.17 ± 0.86[42] in the same units, reporting offsets of 0.156σ and approximately 0.00134σ under those specific comparison conventions. These external numbers are comparison results under the source paper's stated measurement conventions, rather than interchangeable determinations of one identical CPTG observation coordinate. The Hubble bridge therefore evaluates two *derived projections of one native history*; the small endpoint residuals do not, by themselves, constitute new distance measurements. The next test asks a more restrictive question: whether the acoustic branch that produces the first endpoint also reproduces a vector of redshift-dependent observed distance ratios.

5.4 The DESI ruler: carrying the CMB boundary into BAO ratios

The standard-ruler analysis advances beyond endpoint comparisons by evaluating the *shape and normalization* of the DESI distance-ratio vector under the same acoustic projection. Its two upstream observational inputs are the CMB acoustic angle $100\theta_\star^{\mathrm{obs}} = 1.04110 \pm 0.00031$ and the adopted, Planck-derived decoupling coordinate $z_\star = 1089.92 \pm 0.25$[43]. They are comparison inputs, not CPTG-native invariants. With $I_{\mathrm{ac}}(z) = \int_0^z dz'/E_{\mathrm{ac}}(z')$ and $I_\star = 3.213146044210494$, the acoustic ruler is $r_{\mathrm{ac}} = \theta_\star^{\mathrm{obs}}(c/H_{0,\mathrm{ac}})I_\star = 148.621869963\,\mathrm{Mpc}$ in the fully acoustic dimensional convention.[16]

$$\frac{D_M(z)}{r_{\mathrm{ac}}} = \frac{I_{\mathrm{ac}}(z)}{\theta_\star^{\mathrm{obs}} I_\star}, \qquad \frac{D_H(z)}{r_{\mathrm{ac}}} = \frac{1}{\theta_\star^{\mathrm{obs}} I_\star E_{\mathrm{ac}}(z)}. \tag{45}$$

The common dimensional $c/H_{0,\mathrm{ac}}$ factor cancels from these ratios. The controlling fit consequently does not optimize a DESI-specific dimensional ruler or global normalization. The data vector combines the lower-redshift DESI DR2 covariance information[45] with the joint Lyα AP+BAO constraint at $z_{\mathrm{eff}} = 2.33$ and is evaluated in the actual reported observational coordinates:[16]

$$\chi^2_{\mathrm{BAO}} = (\boldsymbol{m}_{\mathrm{CPTG}} - \boldsymbol{d})^T \boldsymbol{C}^{-1} (\boldsymbol{m}_{\mathrm{CPTG}} - \boldsymbol{d}), \tag{46}$$

This yields $\chi^2 = 13.6024620904$ for 13 reported observables ($p = 0.4024238163$ under the source paper's degrees-of-freedom convention). The largest marginal residual is 1.61σ, and the CMB-anchored normalization is 0.779σ from the

conditionally DESI-preferred value. The *unprojected native* branch fails the corresponding source-paper BAO comparison even after a common normalization adjustment. The result is therefore specific to the acoustic projector and the observed CMB boundary. The dimensional $r_{\rm ac}$ above represents the declared acoustic comparison convention; it is not a second independently measured physical ruler.

5.5 The horizon: native containment and common acoustic background

The horizon paper asks whether the physical region that becomes the observed last-scattering surface is contained within the native curvature-transport scale at the same geometric stage. At the source paper's adopted last-scattering phase, $z_\star = 1089.92$, it obtains[17]

$$L_\star = 12.72029287 \text{ Mpc}, \qquad R_C = 23.64711498 \text{ Mpc}, \qquad \Lambda_R \equiv \frac{L_\star}{R_C} = 0.53792156 < 1. \tag{47}$$

The native physical radius and the compared last-scattering radius refer to the same adopted epoch, but they are different quantities from the BAO ruler $r_{\rm ac}$. The paper joins their containment relation to the common geometric phase and acoustic timing projector. The result concerns the *background* geometry: it neither invokes superluminal thermal communication nor supplies the observed CMB map's angular phases. That separates the horizon argument from the map-space reconstruction that follows.

5.6 Planck and WMAP: direct evaluation of a fixed angular envelope

The map study is a visually direct test of the fixed geometric-pi CMB comparison construction. Before the individual map comparisons, its CPTG coordinate row is fixed. A numerical CAMB comparison-coordinate adapter converts that row to an angular-power envelope; a separate fixed Planck control row is passed through the same adapter. The adapter uses explicit comparison-coordinate inputs, including the source paper's $\Omega_{\rm c}h^2$ coordinate, and declared solver assumptions. Those input labels are not promoted to CPTG-native material components. The map-space operation then transports each envelope into the observed masked pseudo-harmonic field:[18, 14]

$$a^M_{\ell m} = a^{\rm obs}_{\ell m}\left(\frac{C^M_\ell}{C^{\rm obs}_\ell}\right)^{1/2}, \quad \ell \geq 2, \qquad T_M(\hat{\boldsymbol{n}}) = \sum_{\ell m} a^M_{\ell m} Y_{\ell m}(\hat{\boldsymbol{n}}). \tag{48}$$

Here M denotes the CPTG or Planck comparison envelope. The observed Planck component-separation map[44] supplies the harmonic phases and the measured pseudo-spectrum $C^{\rm obs}_\ell$, while C^M_ℓ supplies the model-dependent envelope. The construction is therefore a fixed-envelope *comparison-map closure* under observed phases, not a calculation of those phases from unobserved initial data. The reported fitted metric additionally applies the same global amplitude and offset, $T_{M,\rm fit} = A_M \widetilde{T}_M + B_M$, to both comparison envelopes. The raw-map residual is also retained so the reader can distinguish envelope-amplitude mismatch from normalized shape agreement.[18]

Planck/WMAP comparison diagnostic	**Reported value**	**Interpretation**
SMICA observed–CPTG map correlation	0.983434	Shared observed phase scaffold
SMICA raw observed–CPTG residual RMS	26.784740 μK	No global amplitude projection
SMICA globally fitted residual RMS	18.218928 μK	Same amplitude–offset rule
Five Planck component-map mean CPTG–Planck residual-RMS gap	+0.023518 μK	Planck reference ahead
Shuffled-spectrum control mean residual RMS	95.461 μK	Control under same translation

Table 10: Reported CPTG temperature-map comparison values from the CPTG Planck/WMAP study. The temperature-map comparisons use the declared observed harmonic scaffold and common global normalization.

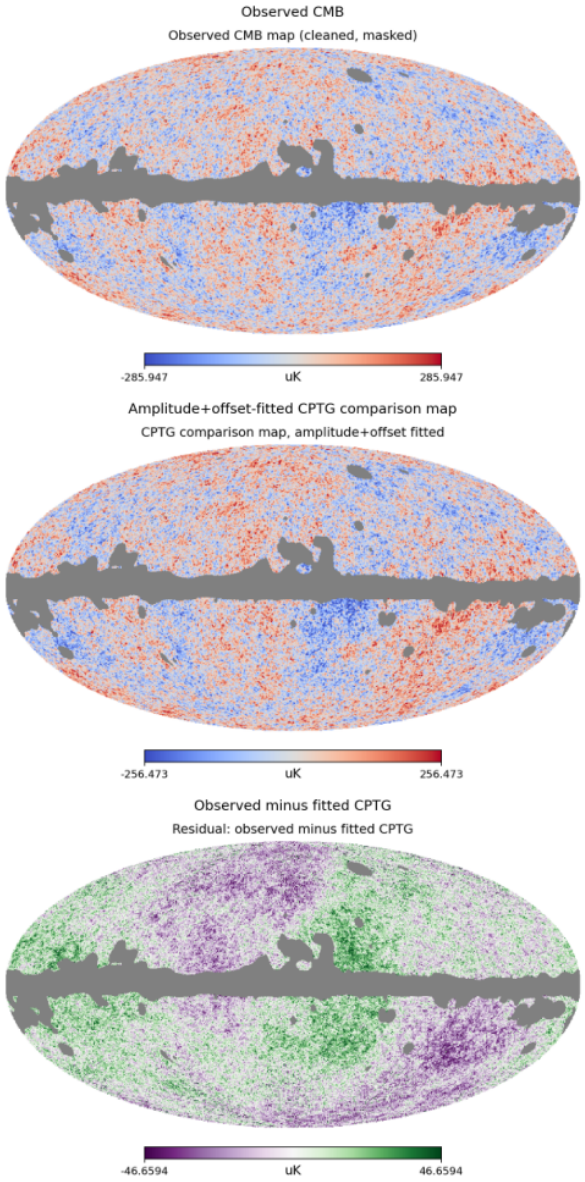


Figure 3: Original figure from the CPTG Planck/WMAP comparison-map study: observed SMICA temperature field, amplitude-plus-offset CPTG envelope transfer using the observed phase scaffold, and their observed-minus-CPTG residual. Reproduced from the original study (Fig. 1), rather than a newly generated CMB sky.

The study carries the fixed CPTG envelope through Planck component maps and splits, WMAP low-ℓ products, smoothing, mask, and null-envelope controls without changing its theory-side map row. Generic flat, power-law, and shuffled controls produce substantially larger residuals under the same observed-phase rule. The high fitted correlation and the fitted residual RMS are mathematically related after the global normalization, so they characterize one agreement rather than independent discoveries. The observed phases, global amplitude projection, and CAMB conversion are visible parts of the reproducible procedure. Once its upstream envelope and observed coordinate are available, constructing the comparison map is a direct numerical operation, without map-by-map theory optimization. No measured runtime advantage over unlike end-to-end pipelines is inferred here.

The progression from Hubble to DESI to the CMB is thus explicit. The Hubble bridge defines comparison endpoints from the native branch; DESI tests the same acoustic projector across a measured redshift-dependent distance vector; native containment addresses the background geometry at last scattering; and the CMB map study tests how a fixed angular envelope is carried into observed map coordinates. The shared theory-side framework is the unifying proposition, while each result retains its own observational inputs, coordinate conventions, and evidentiary meaning.

6 Early Massive Galaxies: Native Phase and Measured Gravitational Organization

The early-galaxy challenge is more specific than the existence of luminous sources at high redshift. Some galaxies observed at high redshift already show population-inferred stellar masses of order $10^{11} M_\odot$, compact structures, and low star-formation activity. Those stellar masses, formation epochs, and quenching histories depend on adopted stellar-population models; quoted statistical errors do not automatically include all initial-mass-function, dust, star-formation-history and distance-systematic uncertainty. A gravitational explanation must distinguish the *cosmological coordinate* used to place an observation, the measured *stellar population and material source*, and the *field required to support its dynamics*. The registered CPTG studies investigate this separation for compact and pressure-supported galaxies observed primarily at $1.2 \lesssim z \lesssim 3.7$ and, in one resolved example, for a two-dimensional stellar-kinematic field. The much higher formation redshifts in the population reconstructions are *inferred formation markers*, not spectroscopic redshifts of those galaxies. They do not calculate the stellar-formation efficiency, gas-cooling history, or detailed quenching mechanism from the gravitational field alone.[30, 31, 32, 33, 37]

6.1 The clock determines the epoch, not an adjustable galaxy force

The native history and acoustic/luminosity projections have been defined in Section 5. Here spectroscopic redshift identifies the stage at which each observed galaxy and its inferred stellar-population markers are placed; the local galaxy-support law remains a separate calculation. For an observed redshift z,[30, 31, 15]

$$a = \frac{1}{1+z}, \qquad \mathcal{N}_\pi = \ln a = -\ln(1+z). \tag{49}$$

The native clock is obtained by integrating Eq. (41), rather than importing the age inferred under the observational source paper's flat-ΛCDM distance convention. This is a change of model-defined cosmic chronology, not an observation of an independent galaxy clock. Similarly, a published stellar mass and radius are translated from their declared luminosity- and angular-distance conventions before they enter the local calculation. In several source studies a neutral conversion uses the arithmetic midpoint of the CPTG acoustic and luminosity *distance coordinates*. That is an explicitly chosen source-unit convention, *not* a fourth cosmological branch or an empirical determination of the true distance. Leading-order distance rescaling of stellar mass does not replace refitting the spectra and stellar populations. Redshift does not multiply an intrinsic galaxy structural acceleration by $H(z)$ or $E(z)$.[30, 31, 33, 37]

The exceptional example is ZF-UDS-7329, also designated PRIMER-EXCELS-55410. Its spectroscopic observation is at $z = 3.1954 \pm 0.0002$, with source-reported $\log_{10}(M_\star/M_\odot) = 11.19 \pm 0.02$ and a population-inferred central formation marker $z_{\rm form} = 14.5^{+8.9}_{-4.2}$ (with a separate quenching marker $z_{\rm quench} = 9.90^{+7.79}_{-3.29}$). The narrow quoted stellar-mass error belongs to the source fit and should not be interpreted as a complete stellar-population systematic budget. The central marker corresponds to $\mathcal{N}_{\pi,\rm form} = -2.74084$ and $t_{\pi,\rm form} = 0.21451\,$Gyr on the registered native clock. At observation, the source study gives $\mathcal{N}_{\pi,\rm obs} = -1.43399$ and $t_{\pi,\rm obs} = 1.66453\,$Gyr. Its average native-phase increment per gigayear is about 6.97 times greater between the *central inferred* formation marker and observation than between observation and the normalized present; using the source's lower 1σ formation-redshift marker changes this ratio to approximately 5.84. This is a comparison of $\Delta \ln a/\Delta t_\pi$ *within one registered clock*, not a claim that atomic, cooling, or stellar-evolution processes run 6.97 times faster, nor a calculation of the stellar mass assembled in that interval. Formation and quenching redshifts are model-inferred population markers with substantial uncertainties, not directly observed clocks.[30]

6.2 From measured stellar support to the local field

The galaxy dynamical reductions then use observed stellar source structure and motions rather than an assumed dark-halo source. In a declared *one-radius pressure-supported estimator*, $a_\star \sim \sigma_\star^2/R_e$ supplies an object-specific support scale. A Sérsic source is deprojected under the adopted spherical or oblate geometry, and the registered CPTG galaxy response gives a field subsequently passed through virial or Jeans observables. This estimator depends on the measured velocity support, and so agreement with a dispersion or virial quantity that entered $a_\star$ is *shared-input closure*, not an independent held-out forecast. The converted virial comparator also shares radius and dispersion inputs, so a small nominal sigma residual alone is not an independent measurement of the gravitational sector. Structural Mode N is computed from the completed radial field and its organization weight, never adjusted to obtain the closure.[30, 31, 33, 5]

The observational distinction among these cases is instructive. XMM-VID1-2075 is a predominantly dispersion-supported slow rotator. Its independently reported $\sigma_{\star,R_e} = 387 \pm 22\,{\rm km\,s^{-1}}$ gives a stringent projected Jeans observable: the source

paper reports $346.48\,\mathrm{km\,s^{-1}}$ for its adopted isotropic stellar-only field and $394.16\,\mathrm{km\,s^{-1}}$ for CPTG. The latter differs from the observed central value by 0.33σ, but the observed dispersion also enters the declared structural-scale estimator. This is a *conditional field consistency* calculation, not a distinct prediction of the same dispersion.[33]

JWST-SUSPENSE 129982 supplies a different test because ordered rotation and stellar pressure are both substantial. Its published effective-radius velocity is $334^{+52}_{-46}\,\mathrm{km\,s^{-1}}$, while the source's central stellar dispersion is $213^{+11}_{-2}\,\mathrm{km\,s^{-1}}$. The CPTG paper constructs three antisymmetric rotation coordinates from the inner public two-dimensional spectrum reaching approximately $0.62R_e$, withholds the published effective-radius rotation from that inner rotation reconstruction, and solves the pressure boundary condition jointly with the baryon-sourced field. Its result $320.9\,\mathrm{km\,s^{-1}}$ is $13.1\,\mathrm{km\,s^{-1}}$ below the published central value. The effective-radius dispersion is still an input to the pressure anchor, through the assumption $\sigma_R(R_e) = 213\,\mathrm{km\,s^{-1}}$. The three inner velocities are reconstructed from public reduced spectroscopy with empirical weights, and the spatial-transfer and one-channel Jeans prescriptions are adopted rather than independently constrained by a complete tensor or covariance analysis. This is a mixed-support cross-check against a withheld *source-inferred* rotation endpoint, not a fully source-only forward forecast.[34]

MQN01 J004131.9–493704 (the "Red Potato") supplies a sixth distinct galaxy. At $z = 3.2496$, the source adopts $\sigma_\star = 268 \pm 20\,\mathrm{km\,s^{-1}}$, a midpoint-converted $R_e \simeq 0.891\,\mathrm{kpc}$, and a pressure-based $a_\star = 2.6126 \times 10^{-9}\,\mathrm{m\,s^{-2}}$, obtaining $N = 3.09498$ on its sampled spherical structural estimator. Its compact quiescent stellar body coexists with a gas-rich environment; the measured-source gravity calculation does not establish the cause of quenching. Its hypothetical present-coordinate Structural Mode values are *fixed-source comparison readouts*, not the observed evolution of the galaxy into a descendant.[37]

Galaxy	z	**Reported CPTG dynamical or structural result**	**Post-solution N**
ZF-UDS-7329	3.1954	$g_{C,e} = (1.498 \pm 0.074) \times 10^{-8}\,\mathrm{m\,s^{-2}}$; one-radius virial comparator $+0.06\sigma$	3.46894 (oblate)
ZF-COS-20115 (integrated)	3.715	$g_{C,e} = 2.1791 \times 10^{-8}\,\mathrm{m\,s^{-2}}$; within 6.4% of converted virial comparator	3.45081 (oblate)
XMM-VID1-2075	3.44927	$\sigma_{\mathrm{ap,C}} = 394.16\,\mathrm{km\,s^{-1}}$ vs. observed 387 ± 22	3.06602 (oblate)
JWST-SUSPENSE 129982	1.249	Mixed-support $V_C(R_e) = 320.9$ vs. $334^{+52}_{-46}\,\mathrm{km\,s^{-1}}$	2.9191
MQN01 J004131.9–493704	3.2496	$\sigma_\star = 268 \pm 20\,\mathrm{km\,s^{-1}}$; one-radius support $a_\star = 2.6126 \times 10^{-9}\,\mathrm{m\,s^{-2}}$	3.09498
NMBS-C7447	1.80	$a_\star = (1.708 \pm 0.613) \times 10^{-9}\,\mathrm{m\,s^{-2}}$; $a_\star/a_\pi = 1.036 \pm 0.372$	3.06465
ZF-COS-20115 (resolved)	3.7135	26 regions; stationary-orbit closure within marginal 3σ criterion	3.43345

Table 11: Selected results from the registered high-redshift and compact-quiescent-galaxy studies. These seven rows cover six distinct galaxies: the two ZF-COS-20115 reductions are not independent targets. The different support estimators, source conversions, observation operators and radial definitions prevent treating the rows as one homogeneous statistical sample. Values of N are object- and reconstruction-specific post-solution measurements; the two ZF-COS-20115 rows describe different reductions of *the same galaxy*, not two galaxies.

6.3 Resolved nonaxisymmetric stellar dynamics: one field, multiple apertures

ZF-COS-20115 has been studied twice at distinct levels of detail. The first study uses its integrated dispersion, size, stellar source, and population history to establish one-radius support and native-phase placement. The later JWST/NIRSpec analysis resolves 26 stellar-kinematic regions of the same galaxy, reporting $z = 3.7135 \pm 0.0002$. Each region supplies its own measured combination of ordered and random support and hence its own $a_{\star,i}$, evaluated through the same registered CPTG response law on a nonaxisymmetric stellar source. The source's structural-radius convention is material: replacing the adopted projected flattening by the stellar-source axis ratio changes the inferred $a_{\star,i}$ by up to 17.67%; circular radii change some values by up to 42.34%. The 26 local field vectors are embedded in a constructed conservative three-dimensional potential, followed by energy-qualified orbit integrations and a projected stellar-kinematic operator. Matching the vectors at these anchors does not establish uniqueness of the potential between or beyond the measured positions.[31, 32]

In the reported 4096-trajectory library, a nonnegative stationary superposition meets the source's *marginal per-bin* 3σ criterion for the 26 measured line-of-sight velocities and second moments at the strictest stated 1% tracer tolerance; the maximum pulls are 2.431σ and 2.879σ. A matched stellar-only control does not meet the all-bin second-moment criterion (3.647σ maximum at that tolerance). The source records $N = 3.433448083$ on its sampled radial estimator and compares it with the earlier oblate $N = 3.45081$, a difference of approximately 0.503%. These are numerical compatibility and structural-continuity findings *within the adopted source geometry, conservative-potential construction, tracer tolerance and fitted orbit weights.* They are *not* blind predictions of the same 26 measurements, because the measurements construct the local structural scales and the orbit weights are optimized against the observed moments. The two measurements of N remain separately reported rather than averaged.[32, 31]

6.4 Early-galaxy results across different dynamical measurements

The NMBS-C7447 analyses expose a particularly important cross-scale ambiguity. At the *observed* redshift $z = 1.80$, both the native-locking and local-structural-invariance reductions recover the same $a_\star/a_\pi = 1.036 \pm 0.372$ and $N = 3.06465$. They diverge only when one imagines relocating the fixed observed object to a different native phase: one prescription keeps the ratio constant, whereas the other keeps the intrinsic galaxy structure and $a_\star$ fixed. Present observations of this one galaxy cannot select between those counterfactual continuation rules. No universal gravitational acceleration–Hubble-rate locking is inferred from the near-unit observed ratio.[35, 36]

Comparison type	**Inputs used in reconstruction**	**Comparison scope**
Native clock and population	Spectroscopic redshift; inferred formation and stellar-population markers	Registered coordinate translation; not an independent measurement of stellar-assembly time
Single-radius gravity (UDS, COS, XMM, MQN)	Measured size, dispersion and adopted stellar source	Shared-input virial, Jeans, or structural closure
SUSPENSE mixed support	Inner reduced-spectrum rotation; effective-radius dispersion anchor	Source-inferred effective-radius rotation withheld from inner rotation fit, but not all kinematic support
Resolved COS and NMBS comparisons	26 local stellar-support bins and fitted orbit weights; or observed state plus counterfactual phase map	Conditional stationary compatibility; unselected competing continuation rules for NMBS

Table 12: Types of observational comparison for the high-redshift examples. Each claim is conditioned on its declared input and source model; entries are not independent constraints on one common fit.

The early-galaxy calculations bring cosmological epoch, stellar sources and kinematic support into a common CPTG analysis. The native clock assigns a geometric phase to the observed and stellar-population-inferred epochs; the registered gravitational reductions then reconstruct the support of massive, compact and quiescent systems. Integrated dispersion, mixed rotational support and resolved stellar-kinematic fields provide increasingly detailed views of gravitational organization, with Structural Mode N recorded after each reconstruction. The two ZF-COS-20115 analyses show how the measured fingerprint can be compared across independent reconstruction procedures for the same object.[30, 32, 33, 35]

7 Primordial Nuclear Geometry and the Lithium Problem

The nuclear-reaction program carries registered CPTG source organization and transport coordinates into a second physical domain. A gravitational object is specified by matter, motion, and a completed spacetime field; a primordial reaction network instead evolves species abundances through thermonuclear kinetics. Their connection lies in inherited geometric comparison coordinates, constrained source directions, and transport prescriptions, with the native nuclear rates and thermodynamic evolution retained. The nuclear papers identify those inherited ingredients explicitly rather than asserting that the spacetime field equation is itself a reaction-rate equation.[2, 8, 19]

7.1 The four-sector source geometry and its conservation law

The registered light-nuclide geometry orders the source coordinates as vertex, bridge, closure, and saturation:

$$(n, p) \longrightarrow D \longrightarrow (T, {}^3\mathrm{He}) \longrightarrow {}^4\mathrm{He}. \tag{50}$$

This diagram identifies organizational sectors, not a literal four-reaction network. At the declared late-BBN light-sector boundary, where surviving free-neutron abundance is negligible, the baryon-normalized state is[8]

$$\boldsymbol{X} = (X_V, X_B, X_C, X_S)^T = \big(Y_H,\, 2Y_D,\, 3(Y_T + Y_{^3\mathrm{He}}),\, 4Y_{^4\mathrm{He}}\big)^T. \tag{51}$$

Its normalization is $\mathbf{1}^T\boldsymbol{X} = 1 - \epsilon_{A>4}$; the reported maximum omitted heavier-nuclide baryon fraction at the six PRIMAT

authority anchors is $2.89372815 \times 10^{-9}$. The corresponding exact ordered-edge operator on zero-sum tangents is

$$\frac{d\boldsymbol{X}}{dx} = \boldsymbol{D}\boldsymbol{j}, \qquad \boldsymbol{D} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{1}^T\boldsymbol{D} = \mathbf{0}^T. \tag{52}$$

Three edge-current directions preserve the baryon sum; one additional charge-constrained polarization direction retains information about the free-nucleon and mass-three mirror composition. The parent source construction writes its internal polarization as $\Pi = \delta P_V = -\delta P_C/3$. Together these form the *rank-four reaction-source coordinate* established for the declared light sector. That source-space result is exact in its stated coordinates, not an assertion that the abundance network itself has only four independent reactions.[8]

The scalar source response inherits the acoustic baryon-density coordinate η_{10}^{ac} and the geometric transport factor $G_T = (3 - \pi/100)/\pi$. The first- and second-order coefficients are[2, 8]

$$\begin{aligned} J_{\mathrm{CPTG}}(\eta_{10}) &= \frac{1}{\sqrt{8}} \left(\frac{\eta_{10}}{\eta_{10}^{\mathrm{ac}}}\right)^{1/2}, \\ K_{\mathrm{CPTG}}(\eta_{10}) &= -\frac{3}{16}\frac{\eta_{10}}{\eta_{10}^{\mathrm{ac}}} G_T^{-1/3} = -\frac{3}{2} J_{\mathrm{CPTG}}(\eta_{10})^2 G_T^{-1/3}. \end{aligned} \tag{53}$$

Thus the second-order coefficient is constrained by the first rather than independently selected for each reaction. The fixed reference values are $\eta_{10}^{\mathrm{ac}} = 6.17038016771$ and $G_T = 0.9449296585513721$. Their transported BBN comparison coordinate is $\eta_{10}^{\mathrm{BBN}} = \eta_{10}^{\mathrm{ac}}\sqrt{G_T} \simeq 5.998071834744$; it must be distinguished from the local η_{10} at which a source-response operator is evaluated. Network rates, cross sections, thermal histories, and Jacobians remain defined within the native nuclear calculations.[2, 8]

7.2 From the commissioned capture channel to independent network checks

The commissioned $D(p,\gamma)^3\mathrm{He}$ construction connects a fixed nuclear source amplitude with an astrophysical S-factor and rate interface. Its reported zero-energy value is $S_{12}^{\mathrm{CPTG}}(0) = 0.2145003969\,\mathrm{eV\,b}$, compared with the cited Solar Fusion II/LUNA extrapolation $0.216 \pm 0.010\,\mathrm{eV\,b}$, giving the stated signed difference -0.149960σ under that comparison. The dimensional S-factor requires an explicitly defined normalization of the dimensionless source intensity; matching one measurement does not establish every energy-dependent capture observable.[19]

The four-sector study next tests whether fixed source directions survive propagation through native nuclear networks. Its full PRIMAT campaign processes 428 native reactions at six baryon-density anchors. Within that larger campaign, the designated primary source/Jacobian series reports **48/48** passing resolved reaction–anchor checks, a worst endpoint-vector discrepancy of 0.0367629%, and minimum direction cosine 0.99999999209394. In the common Reaction-20 logarithmic-response coordinate $(D/H, {}^3\mathrm{He}/H, {}^7\mathrm{Li}/H)$, the median directions from PRIMAT, PArthENoPE, and PRyMordial have pairwise cosines above 0.99998. The agreement is in the declared response direction, while the numerical response matrices remain native to each code.[8]

7.3 The live mass-seven transport and its abundance result

The cosmological lithium discrepancy concerns surviving mass seven, much of which is initially produced as $^7\mathrm{Be}$ and subsequently contributes to $^7\mathrm{Li}$. The nuclear state therefore includes $Y_7 = Y_{^7\mathrm{Li}} + Y_{^7\mathrm{Be}}$. Its CPTG abundance-transport prescription is *separate* from the four-sector light-source simplex. Both live channels receive the same extra sink during the registered network-evolution window:

$$\left.\frac{dY_i}{dt}\right|_{\mathrm{CPTG}} = \left.\frac{dY_i}{dt}\right|_{\mathrm{network}} - \Gamma_{7,\mathrm{trans}}(t) Y_i, \quad i \in \{^7\mathrm{Li}, {}^7\mathrm{Be}\}, \qquad \tau_7 = \int \Gamma_{7,\mathrm{trans}}(t)\,dt = \ln \pi. \tag{54}$$

For an isolated channel, $e^{-\tau_7} = 1/\pi = 0.3183098862$. In the coupled network, reactions continue to produce and destroy mass-seven nuclei while the transport term operates, so the final ratio is an output rather than an imposed number. The PRIMAT v0.3.2 campaign freezes the optical depth and a log-cosine temperature window from 0.0175 to 0.0040 MeV before evaluating the lithium endpoint. The two sink terms are included in the live abundance equations and the analytic Jacobian.[20]

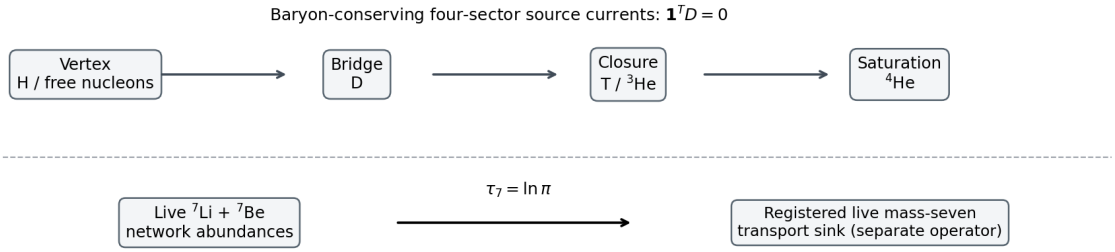


Figure 4: Two separately defined nuclear operations. The four-sector $\boldsymbol{Dj}$ currents conserve the declared baryon sum in their light-source coordinates. The registered mass-seven optical-depth term acts within the live ^{7}Li/^{7}Be equations, with the same network otherwise retained. This diagram does not assign an unmeasured receiving nuclide to the transport sink.

The distinction from the four-sector theorem is physical as well as mathematical: the mass-seven law is an *effective abundance-transport sink*, not one of the three baryon-preserving incidence currents or an identified conventional nuclear reaction. Its two displayed sink terms do not, by themselves, specify a receiving species or a closed microscopic account of the transported material. Conservation of the separate four-sector source operator must not be claimed as a proof of such an account for the mass-seven channel.[2, 8, 20]

PRIMAT output / control	**Gate off**	**Gate on**
^{7}Li/H	$5.2735647904 \times 10^{-10}$	$1.6789263563 \times 10^{-10}$
Live Y_7 survival	—	0.3183664984
D/H relative gate effect	—	3.91×10^{-9}
^{4}He mass fraction relative gate effect	—	3.86×10^{-12}
^{3}He/H relative gate effect	—	5.59×10^{-10}

Table 13: Reported network-native mass-seven transport result and non-mass-seven controls from the dedicated lithium study. The survival ratio differs by 0.0177852% from $1/\pi$ under the paper's relative-error convention. The reported control differences refer to gate on versus gate off at the same transported comparison inputs.

The discrepancy and its resolution can now be stated in the same observational coordinate. Against the declared stellar-lithium comparison anchor $(1.45 \pm 0.25) \times 10^{-10}$[20, 46], the ungated PRIMAT result lies $+15.2943\sigma$ above the anchor and the transported result $+0.915705\sigma$ above it. The anchor was withheld from the transport normalization and used only to describe the outcome. The reported **23/23** preregistered checks cover implementation, numerical survival, and preservation of the non-mass-seven controls; they are not 23 independent abundance measurements.[20]

The experiment uses a frozen transported baryon-and-radiation comparison tuple while retaining PRIMAT's native BBN background. Numerical agreement with the adopted radiation coordinate checks the implementation of that input rather than measuring the coordinate anew. The stellar lithium anchor also carries atmospheric, temperature-scale, and metallicity systematics. Accordingly, the supported result is selective live-network mass-seven reduction at the declared CPTG comparison state, with the ordinary light-element yields essentially unchanged under the stated tests.[20]

7.4 How far the source construction has been carried

The computational companion combines a frozen PRIMAT native inventory through $A = 23$ with a separate prescribed-bath reduced reaction graph above that boundary. It records modeled structural reachability through $A = 119$ at six baryon-density anchors, with $A = 5$ explicitly marked as a physical gap, and a qualified reachable frontier at $A = 338$ under its selected external graph. The eligible $A = 339$ sector is disconnected in that topology. These results establish algorithmic reachability and positive *modeled* support; they are not a coupled native-PRIMAT calculation of primordial heavy-element abundances or an extension of the exact light-sector rank-four theorem to every heavy nucleus.[23, 8]

The nuclear connection to the wider CPTG framework is explicit: the fixed CPTG comparison coordinates supply the

nuclear scalar coefficients; baryon and charge constraints organize the four-sector source; native reaction kinetics evolve that source into abundances; and a separate, inherited mass-seven optical depth changes survival within the live network. This is a common source–response–transport architecture with distinct physical realizations. The fixed geometric coordinates, conserved nuclear-source organization, native reaction kinetics, and registered transport law form a reproducible sequence from source construction to primordial abundances.[2, 8, 20]

8 Finite Black Holes, Compact Mass, and the Structure of Time

The compact-object realization exposes the physical distinction at the center of CPTG's geometric-mass interpretation. The exterior gravitational mass and angular momentum characterize the observable spacetime; they are not a direct census of the localized physical source within its interior. CPTG retains the Einstein tensor and ordinary stress–energy throughout, and recovers the Schwarzschild/Kerr exterior where the additional local geometric response has completed. The registered compact construction then distinguishes source and geometric contributions to that *same* conserved exterior charge, without introducing an additional material halo. The finite stationary profile is a specified admissible realization, not a uniquely derived general rotating interior or a model of the preceding stellar collapse.[7, 2]

8.1 A finite interior with an unchanged exterior charge

The minimum registered spherical continuation uses the normalized coordinate $x = r/r_m$, with matching radius $r_m = GM_\infty/c^2$, and the analytic profile[7]

$$F(x) = \frac{1}{8}\big(63x^5 - 90x^7 + 35x^9\big), \qquad 0 \le x \le 1, \qquad F(1) = 1, \quad F'(1) = F''(1) = 0. \tag{55}$$

The profile is monotone inside the matching surface because its exact derivative is

$$F'(x) = \frac{315}{8}x^4(1 - x^2)^2 \ge 0 \qquad (0 \le x \le 1). \tag{56}$$

It interpolates from $F(0) = 0$ to $F(1) = 1$ without a negative enclosed-mass gradient. The profile behaves as $F(x) = (63/8)x^5 + O(x^7)$ near the center. More explicitly, the *total Misner–Sharp mass-equivalent* scales as $m_{\rm MS} \sim r^5$, so its effective spherical radial density $m'_{\rm MS}/(4\pi r^2)$ is proportional to r^2 near the origin. This is a regularity check on the registered spherical mass function, not a claim that the physical material density is separately determined by the exterior observations. The normalized curvature carrier $U(x) = F(x)/x^2$ also tends to zero as x^3. The source paper embeds this profile in its curvature-bounded finite metric; these mass-function limits alone do not prove regularity for an arbitrary rotating solution.

At the matching boundary, the enclosed gravitational mass and its first two radial derivatives meet the constant-mass exterior continuously, suppressing the boundary stress generators of the transport and polarization channels. The matching radius is a property of this finite realization: it is neither a universal activation threshold nor a replacement of General Relativity at a horizon. For a subextremal Kerr exterior, the ordinary outer horizon lies outside r_m.[7]

The spherical enclosed mass is identified through the Misner–Sharp construction, $m_{\rm MS}(r) = M_\infty F(x)$. In the registered source–transport–polarization decomposition,[7]

$$\begin{aligned} M_c(r) &= M_c F, \\ M_{\rm tr}(r) &= M_{\rm geom}F(1 - F), \qquad M_c(r) + M_{\rm tr}(r) + M_{\rm pol}(r) = M_\infty F(x). \\ M_{\rm pol}(r) &= M_{\rm geom}F^2, \end{aligned} \tag{57}$$

At $F = 1$, the transport channel vanishes while the integrated geometric contribution remains in the polarization channel: $M_{\rm pol}(r_m) = M_{\rm geom}$. Thus a completed geometric contribution may remain part of the conserved exterior mass even after the local response gradients have diminished and the exterior metric takes its familiar GR form. In the spherical sector this is an explicit enclosed-mass accounting identity, not by itself a measurement of three separately resolved stress tensors. The source defines M_c as the compact-state localized-source contribution; it is not automatically identical to the progenitor stellar mass or a separately observed microscopic rest mass. The rotating application described below is a *conditional matching construction*, not an independently solved general axisymmetric interior from the parent action.[7]

8.2 The same mass–spin handoff for different compact states

The matched exterior state $(M_\infty, \hat{a}_\infty)$ determines a *conditional* compact source–geometry decomposition: the finite model identifies one matching radius and retains the same angular momentum in its exterior and localized-source coordinates. These mass and continuum-spin inputs are themselves model-dependent, and the derived M_c is not a new direct measurement of compact material. Let $q = |\hat{a}_\infty|$ and $A_{\rm pol} = M_\infty/M_c$. The registered handoff combines the *common finite-to-Kerr matching radius* with angular-momentum conservation. In the source paper's dimensionless compact spin coordinate,[7]

$$r_m = \frac{GM_\infty}{c^2} = \frac{GM_c}{c^2}\left(1 + \sqrt{1 - \hat{a}_c^2}\right), \qquad \hat{a}_c = A_{\rm pol}^2 \hat{a}_\infty. \tag{58}$$

Using $M_\infty = A_{\rm pol} M_c$ gives $A_{\rm pol} = 1 + \sqrt{1 - A_{\rm pol}^4 q^2}$, which on the physical branch reduces to the fixed compact cubic:[7]

$$q^2 A_{\rm pol}^3 + A_{\rm pol} - 2 = 0, \qquad 1 \le A_{\rm pol} \le 2, \qquad M_c = \frac{M_\infty}{A_{\rm pol}}, \quad M_{\rm geom} = M_\infty - M_c. \tag{59}$$

The physical root follows from *both* the stipulated radius equality and conserved angular momentum; no second mass-split parameter is assigned object by object. At fixed exterior spin magnitude the fractional split is independent of overall mass, falling from one half geometric share at zero exterior spin to zero at extremal spin. This is the registered matching-boundary prescription, not a uniqueness theorem for arbitrary rotating interiors.[7]

The spin dependence of the geometric share can be written explicitly as

$$f_{\rm geom}^{\rm BH}(q) = 1 - \frac{1}{A_{\rm pol}(q)}, \qquad A_{\rm pol}(0) = 2, \quad A_{\rm pol}(1) = 1. \tag{60}$$

These are endpoints of the conditional cubic, not additional source-mass measurements.[7]

Conditional compact quantity	**M33 X-7**	**LMC X-3**
M_∞ $(M_\odot)$	15.65	6.98
Exterior spin magnitude q	0.84	0.21
$A_{\rm pol}$	1.088925987	1.759699717
M_c $(M_\odot)$	14.3719593	3.9665859
$M_{\rm geom}$ $(M_\odot)$	1.2780407	3.0134141
Geometric share $M_{\rm geom}/M_\infty$	8.1664%	43.1721%

Table 14: Two source-reported central mass–spin branches evaluated with exactly the same compact cubic. The input mass and spin measurements have model-dependent uncertainty and are not refitted by CPTG. This compact exterior-charge decomposition is distinct from the galactic spherical-reference mass ledgers of Section 2.

The finite profile replaces the infinite-curvature singular endpoint within the registered stationary construction. The contrasting mass shares illustrate the conditional compact mapping: a smaller geometric share for one compact state and a larger share for another follow from different *adopted exterior spin branches* under the same relation, not from adding or removing material to improve a fit. The comparison must retain the mass–spin pairs used in the source studies; later independently published mass or spin measurements cannot be combined into an invented joint measurement.[7]

8.3 The compact mass gap and the missing-source accounting problem

The mass–spin relation changes what an inferred black-hole mass measures. For the physical root, the fraction $M_c/M_\infty = 1/A_{\rm pol}$ rises monotonically from $1/2$ at $q = 0$ to unity at $q = 1$, while $M_{\rm geom}/M_\infty$ decreases from $1/2$ to zero. Thus the same measured exterior mass can correspond to a smaller localized compact-source contribution, and that contribution depends on the measured exterior spin. The source mass is calculated from the exterior state rather than assigned by an additional hidden-matter component.[7, 3]

This gives CPTG a concrete interpretation of the conventional lower compact-object mass-gap problem. A mass gap classified by exterior gravitational mass is not automatically the same gap in the localized source-mass distribution:

the registered mapping moves each individual state to its own M_c. It preserves the observed exterior masses and spins while changing their internal mass accounting; the GWTC-5 population transformation illustrates how the inferred compact-source distribution shifts under the fixed relation. In this compact sense it also disentangles a "missing baryon" or missing-source inference: the difference $M_\infty - M_c$ is a geometric contribution to gravitational charge, not material that must be invented to reproduce that charge. This compact accounting is distinct from a census of diffuse cosmic baryons.[7, 21, 3]

8.4 Structural organization locates internal clock and rotation readouts

The mode is a post-solution measurement on the finite compact field, not a parameter used to determine the matching cubic.[7, 5]

For the finite compact continuation, the structural operator is constructed from the normalized solved mass profile and an equal-measure rotating coordinate. The retained radial shape is $F(x) = (63x^5 - 90x^7 + 35x^9)/8$ for $0 \le x = r/r_m \le 1$. Define $U(x) = F(x)/x^2$ (with its regular limiting behavior at $x = 0$). The registered compact carrier, weight and equal-measure coordinate are[5, 7]

$$C_{N,\mathrm{BH}}(x) = U^2(x) + [U'(x)]^2 + 2\left[\frac{U(x)}{x}\right]^2, \qquad w_{N,\mathrm{BH}}(x) = \left|\frac{dC_{N,\mathrm{BH}}}{dx}\right|, \tag{61}$$

$$\xi(x,q) = \left[\frac{x(x^2+q^2)}{1+q^2}\right]^{1/3}. \tag{62}$$

The compact organization length and mode are then

$$\lambda_{N,\mathrm{BH}} = r_m \frac{\displaystyle\int_0^1 \xi(x,q) w_{N,\mathrm{BH}}(x)\,dx}{\displaystyle\int_0^1 w_{N,\mathrm{BH}}(x)\,dx}, \qquad N_{\mathrm{BH}} = \frac{r_m}{\lambda_{N,\mathrm{BH}}}. \tag{63}$$

The spherical limit in the registered realization is $N_{\mathrm{BH}}(0) = 1.912233171442$. The mode also locates an interior organization shell $r_N = x_N r_m$ through $x_N(x_N^2 + q^2) = (1+q^2)/N_{\mathrm{BH}}^3$. Lapse and frame-dragging readouts are evaluated on the corresponding completed geometry; neither those readouts nor N_{BH} is fed backward to select the compact mass–spin closure.[5]

At that shell, the registered Kerr-compatible finite-profile metric fixes the lapse and zero-angular-momentum-observer (ZAMO) frame-dragging rate. Before interpreting those clock values, the location relative to the model's inner trapping surface must be identified. With the dimensionless functions[7]

$$\begin{gathered} \delta = x^2 - 2xF(x) + q^2, \qquad \sigma = x^2 + q^2\cos^2\theta, \\ \mathcal{A} = (x^2+q^2)^2 - q^2\delta\sin^2\theta, \\ \alpha(x,\theta;q) = \sqrt{\frac{\delta\sigma}{\mathcal{A}}}, \qquad \omega(x,\theta;q) = \frac{c}{r_m}\frac{2qxF(x)}{\mathcal{A}}, \end{gathered} \tag{64}$$

the lapse gives a local stationary-coordinate proper-time factor, while ω measures geometric frame dragging rather than material rotation. The inner trapping surface is the subextremal root x_- of $\delta(x_-, q) = 0$ in $0 < x_- < 1$. For the two benchmark profiles, $x_N < x_- < 1$: the organization shell lies *inside the inner trapping surface*, while the ordinary exterior Kerr horizon lies outside r_m. This placement is essential for interpreting the lapse; the shell is not in the region between the two trapping surfaces. At $q = 1$, the corresponding root reaches the matching endpoint and must be treated as a limit.[7]

Let an overbar denote the source-defined $\sigma\sin\theta$-weighted angular mean over polar angle. For an interval of 24 source-local asymptotic exterior *coordinate* hours, the formal stationary-shell readouts are $T_{24,N} = 24\,\overline{\alpha}(x_N;q)$ h and $\overline{\omega}_N = (c/r_m)\widehat{\omega}_N(q)$. These are calculated local geometric quantities in the specified coordinate construction, *not* time delays observed in electromagnetic signals arriving from behind an event horizon and not a prediction that a physical observer can remain at

that shell without additional dynamical conditions. Structural Mode selects the shell; the metric supplies its clock rate and local rotational geometry.[7]

Post-solution quantity	**M33 X-7**	**LMC X-3**
Compact Structural Mode N_{BH}	1.493871633444	1.831967029641
Organization radius r_N (km)	12.0792508	5.43440975
Inner trapping coordinate x_-	0.815809395077	0.597095316533
Local shell-clock readout in 24 coordinate hours	19.5877337 h	13.1737826 h
Geometric $\overline{\omega}_N$ (rad s^{-1})	3683.246	13248.618

Table 15: Source-reported structural and local geometric readouts, including the inner trapping coordinate, for two conditional exterior states. Both have $x_N < x_- < 1$. These quantities are calculated after the mass–spin handoff and are not direct astronomical interior-clock measurements.

This provides a model-defined local structure-of-time calculation: the geometry supplies the lapse and frame dragging, while Structural Mode locates the shell at which they are evaluated. The finite continuation does not eliminate ordinary exterior Kerr orbital or horizon dynamics. Nor does this post-solution readout establish the interior clocks observationally, a complete collapse history, or the uniqueness of the rotating continuation.[7, 5]

8.5 Singular endpoints, geodesic continuation, and black-hole information

The registered finite construction replaces the classical divergent compact endpoint with a smooth central mass profile and a continuous exterior match. Its $F(x) \sim x^5$ center gives a vanishing radial curvature carrier and a finite reduced curvature description; the completed stationary model therefore has no infinite-curvature point serving as a physical terminus. The classical singularity theorems establish geodesic incompleteness under their specified assumptions. CPTG changes the strong-field continuation by retaining the Einstein sector while adding polarization and transport, so the singular classical endpoint is no longer the prescribed completion of the compact state. A statement about the global completeness of every causal geodesic requires the full matched spacetime and is distinct from the regular-center result.[7, 24, 2]

This addresses the singularity-dependent part of the black-hole information paradox directly. In the CPTG compact construction, information cannot be declared destroyed by arrival at an infinite-curvature singularity, because that physical endpoint is absent. The exterior Schwarzschild or Kerr horizon remains, so the ordinary semiclassical Hawking process is not discarded by removing the singularity. Information transport and evaporation must instead be described on the finite source–polarization–transport geometry. The physical advance is a finite continuation with no singular information sink; the detailed quantum evolution is not supplied by the stationary mass profile alone.[7]

8.6 Catalog-wide compact consistency and a withheld timing case

The GWTC-5 compact-population study applies the same native observation-coordinate reconstruction and compact handoff to a declared 259-event binary-black-hole set. Its source registry retains 288 release-label realizations and 260 unique numerical streams, corresponding to 520 component states and 7,626,239 posterior draws. All declared events satisfy the study's numerical closure gates; the maximum compact cubic residual is 1.332×10^{-15}. The catalog's component Structural Modes remain object-specific rather than being combined into a single physical average. A separate selection-corrected population calculation propagates 8,200 hyperposterior rows with 512 joint realizations per row. These are extensive checks of numerical and posterior consistency of the registered compact mapping. Their millions of correlated Monte Carlo samples are not millions of independent tests of compact-interior physics; the released observations remain the exterior-state authority.[21, 7]

The off-nuclear candidate 3XMM J215022.4−055108 supplies a distinct comparison. An independently fitted, astrophysical-model-dependent multi-epoch X-ray disk radius was constructed without the observation containing its reported 11.88 ± 0.38 mHz quasi-periodicity. With $z = 0.055$ this gives 12.5334 ± 0.4009 mHz in the source-frame timing coordinate. Conditional on identifying that signal with Lense–Thirring precession at the innermost stable circular orbit, the published radius and timing condition select $q \simeq 0.5005$ and $M_\infty \simeq 2.7711 \times 10^4 M_\odot$. Only then does the unchanged CPTG compact handoff yield $M_c \simeq 2.0314 \times 10^4 M_\odot$ and $M_{\mathrm{geom}} \simeq 7.397 \times 10^3 M_\odot$, or a 26.69% geometric share. The retrospective withheld-input timing closure tests a retained Kerr exterior interpretation conditional on the selected precession mechanism; the CPTG internal mass and clock quantities are downstream mappings, not direct measurements of its finite interior. It is not an independent observational discrimination of CPTG from a retained Kerr exterior by itself.[22, 7]

The compact section therefore links three meanings of physical structure without merging them: conserved exterior gravitational charge, a finite source–polarization–transport interior, and post-solution measurements of organization and time. It also sharpens the difference between a completed finite compact construction and an unrestricted claim about every possible rotating geometry. The displayed registered finite-profile realization is an explicit candidate construction in which an infinite-curvature interior is not assumed as the physical endpoint, while the familiar GR exterior is matched continuously. What remains unestablished here is the unique rotating finite solution of the full coupled action and a dynamical formation history. Those are distinct questions from the algebraic mass closure and the post-solution internal readouts audited in this section.[7, 2]

9 Vanishing Stars and the Formation of Hidden Compact Remnants

A massive star can leave a compact remnant without announcing its death through an ordinary bright supernova. Such an event is sought by first identifying the star, following a weak outburst or prolonged fading, and verifying its subsequent optical absence. This observational channel is especially relevant to the preceding compact chapter: CPTG permits a finite-curvature compact continuation behind a GR-controlled trapped exterior, whereas stellar luminosity, dust production, weak ejecta, and fallback are determined by additional astrophysical processes. Two existing CPTG source studies evaluate this possibility for NGC 6946-BH1 and M31-2014-DS1. Neither treats optical disappearance as direct access to the remnant interior.[24, 25, 7]

9.1 Two observed disappearance sequences, two levels of certainty

NGC 6946-BH1 has a known luminous red-supergiant progenitor, a weak 2009 transient, persistent optical disappearance, and a late red mid-infrared remnant. The cited JWST reanalysis separates four neighboring near-infrared stars from the central remnant. Its *SED-model* estimates, with source-reported 95% marginal confidence limits, are $\log_{10}(L_{\rm rem}/L_\odot) = 4.727^{+0.026}_{-0.025}$ and $\log_{10}(L_{\rm prog}/L_\odot) = 5.632^{+0.023}_{-0.018}$. Their central values define a luminosity diagnostic, not a directly observed interior property or a CPTG fit parameter:[24, 26]

$$R_{L,6946} = \frac{L_{\rm rem}}{L_{\rm prog}} = 10^{4.727-5.632} = 10^{-0.905} = 0.1245 \quad \text{(rounded)}. \tag{65}$$

The *central fitted* remnant luminosity is therefore approximately one eighth of the fitted progenitor luminosity. The individual luminosity uncertainties are supplied by the source, but their joint covariance is not used here; the central-value ratio is not presented as a separately measured, zero-uncertainty quantity. Long-baseline LBT observations *through September 2025*, as reported by the source, find no ordinary optical recovery. The red dust environment, weak-ejection inference, and Chandra nondetection strengthen the overall quiet-collapse case, but a red remnant alone does not prove a trapped compact endpoint.[24]

M31-2014-DS1 is a complementary and more disputed example. Its hydrogen-depleted supergiant progenitor brightened in the mid-infrared beginning in 2014, then became undetectable in the optical and faded strongly in total luminosity. JWST identifies an extremely red dust- and molecule-rich remnant; Chandra has not detected a corresponding X-ray source. The De et al. source-model branch uses $\log_{10}(L_{\rm rem}/L_\odot) \simeq 3.88$ and an *approximate* progenitor reference $\log_{10}(L_{\rm prog}/L_\odot) \simeq 5$, giving a branch-specific central ratio,[25, 27, 28]

$$R_{L,\mathrm{M31,De}} \simeq 10^{3.88-5} = 0.0759. \tag{66}$$

The separate Beasor et al. interpretation estimates the current infrared source at roughly 7–25% of the progenitor luminosity and cautions that anisotropic dust reprocessing can conceal additional intrinsic light. This is a competing source-model interpretation, not a statistical confidence interval around the De et al. ratio. Published interpretations therefore include failed explosion with fallback to a black hole, a dust-enshrouded survivor or stellar merger, and a binary-interaction alternative. The fading and the persistent infrared source are observations; their inferred bolometric ratios depend on the respective dust and SED models. Terminal collapse is *not uniquely established.*[25, 28, 29]

Observable / inference	NGC 6946-BH1	M31-2014-DS1
Stellar history	Luminous red supergiant; weak 2009 transient and optical disappearance	Hydrogen-depleted supergiant; 2014 infrared brightening and later optical disappearance
Published late light	Central SED ratio $R_L = 0.1245$ from JWST reanalysis; individual luminosities have 95% intervals	$R_L \simeq 0.0759$ on De et al. branch; 7–25% with asymmetric-dust qualifications in Beasor et al.
Circumstellar state	Red mid-infrared remnant and modeled weak dusty ejecta	Strong molecular and silicate-dust features; weak molecular-gas ejection
Interpretive status	Strong quiet-collapse candidate; hidden endpoint not directly resolved	Terminal collapse contested against survivor, merger, and binary interpretations

Table 16: Observed phenomena and model-dependent interpretations in the two source studies. The luminosity ratios are central values from nonidentical SED/dust models, not jointly comparable mass fractions or measurements of an interior geometry.

9.2 Weak ejecta and X-ray limits constrain the external event

The NGC 6946-BH1 silicate-dust-shell analysis quoted in its source gives a characteristic ejecta speed near $60\,\mathrm{km\,s^{-1}}$, and model-dependent lower scales $M_e \gtrsim 0.08M_\odot$ and $K_e \gtrsim 3 \times 10^{45}\,\mathrm{erg}$. Its Chandra nondetection is reported with an approximate $1{,}500L_\odot$ luminosity upper limit under the specified power-law spectral assumption. The accompanying $10M_\odot$ Eddington normalization is a reference used in the observational paper, *not* a measured CPTG remnant mass.[24]

For M31-2014-DS1, the source study reports molecular gas near $0.1M_\odot$ moving at approximately $100\,\mathrm{km\,s^{-1}}$, with a characteristic dust structure of 40–200 au. The kinetic-energy scale derived from those inputs is[25, 28]

$$E_{\mathrm{gas}} \simeq \frac{1}{2}(0.1M_\odot)(100\,\mathrm{km\,s^{-1}})^2 \simeq 9.94 \times 10^{45}\,\mathrm{erg}. \tag{67}$$

The same source quotes an absorption- and spectrum-dependent Chandra limit of $L_X \lesssim 1.5 \times 10^{35}\,\mathrm{erg\,s^{-1}}$ in the 0.5–7 keV band. The dust radius and kinetic energy constrain the history of circumstellar ejecta; the X-ray limits constrain exposed accretion and obscuration. Neither measurement determines the interior CPTG polarization or transport response. A nondetection can result from weak accretion, radiative inefficiency, obscuration, or a surviving system, so it is not a unique signature of finite compact curvature.[25, 24]

9.3 The finite structural-collapse interpretation

The compact field construction of Section 8 makes the theoretical implication specific. Ordinary General Relativity remains a continuous part of CPTG and controls the exterior, trapping horizon, and orbital domain. The registered compact framework retains only finite-curvature *admissible* interior branches in its stationary construction; neither vanishing-star study evolves a collapsing progenitor into a unique completed spacetime. For the *reduced compact-field* regular-center check, the source papers write $\Phi(r) = \Phi_0 + \alpha_c r^2 + O(r^4)$ and use the compact field invariant $C_c = g_j g_j + \kappa_c^2(\partial_i g_j)(\partial_i g_j)$. The conditional central limit is[24, 25]

$$C_c(0) = 12\kappa_c^2\alpha_c^2 < \infty \qquad \text{for finite } \kappa_c, \alpha_c. \tag{68}$$

Here C_c is the retained compact-response *field-gradient invariant*, evaluated in the local reduced construction; it is not the four-dimensional Kretschmann scalar. Its boundedness follows from the *assumed regular-center expansion*. It does not by itself establish regularity of every curvature scalar, global metric matching, dynamical stability, or the time-dependent collapse solution. Those require the completed compact field and boundary conditions discussed in Section 8. The existing stationary construction therefore identifies an admissible finite continuation; the radiation-hydrodynamic route into it remains uncalculated for these two events.[24, 25, 7]

Crucially, neither vanishing-star paper fits a remnant spin, a dynamical exterior gravitational mass, M_c, M_{geom}, A_{pol}, compact $a_\star$, or Structural Mode N. In particular, the approximately $5M_\odot$ compact endpoint in one M31 failed-supernova interpretation is a *stellar-collapse model output*, not the independent exterior mass–spin pair needed for the cubic in Eq. (59). Substituting it into that relation would manufacture a precision unsupported by the observations. The absence of a luminous supernova is compatible with the CPTG admissible compact branch, but it is also compatible with conventional failed-supernova physics; the electromagnetic histories alone do not distinguish finite and classical hidden interiors.[25, 24]

9.4 Astrophysical significance of the disappearance events

The two disappearance sequences provide empirical settings for the quiet formation of compact remnants. NGC 6946-BH1 offers a luminous identified progenitor, a weak transient, long optical absence and a red infrared remnant; M31-2014-DS1 adds molecular ejecta, a dust-rich environment and a range of stellar-endpoint interpretations. Their light curves, dust emission and ejecta energetics constrain the exterior astrophysical event. The source-reported LBT failed-supernova fraction $f = 0.16^{+0.23}_{-0.12}$ (90% confidence) illustrates the emerging population study of these events. Continued multi-epoch spectroscopy and imaging refine their remnant classification as circumstellar material evolves. The registered CPTG finite-state construction supplies the geometric description relevant to a compact endpoint once its physical exterior state is established.[24, 25, 7]

The compact localized-source/exterior-charge split is not the same definition as a galaxy's radius-dependent mass equivalent. Both nevertheless bring the same interpretive question into focus: when gravity is inferred, what establishes that the field's entire additional response must correspond to another material constituent? This is the question addressed next.

10 A Reckoning for the Dark Matter Paradigm

The dark-matter hypothesis began with a genuine physical problem: the gravitational fields inferred from motions, lensing and the large-scale distribution of structure cannot generally be identified with the fields attributed to the directly observed stars and gas under the usual material accounting. The observations are real. The additional claim—that their unexplained gravitational contribution *must* be a second, nonluminous material population—is an interpretation of those observations. It is this identification, rather than the observations themselves, that CPTG challenges.[1, 4, 2]

10.1 Where the dark-matter interpretation fails to close the physical explanation

Cold dark matter has been an effective component of the ΛCDM description, including its fits to cosmic microwave background anisotropies and the development of large-scale structure. That empirical usefulness is important, but a successful gravitational mass budget is not an independent identification of the proposed material constituent. The inferred contribution has been assigned density, halo structure and clustering properties without a confirmed particle identity. Searches constrain candidate masses and interactions rather than supplying a verified microscopic source for the inferred cosmic component. Even the unusual interaction reported by LUX–ZEPLIN in 2026 was explicitly below the threshold for a particle discovery. This distinction marks the outstanding *physical-explanation failure* of the dark-matter hypothesis as a material account; it does not mean that all ΛCDM observational fits have failed.[43, 48]

At galaxy scale, the usual cold-halo picture also faces well-documented tensions between simple collisionless halo expectations and the observed organization of luminous systems: central density cores rather than steep cusps in many dwarfs, variation in the shapes of rotation curves at comparable outer velocities, and the abundance and internal structure of faint satellites. Galaxy formation, stellar feedback, selection effects and more elaborate dark-sector models can mitigate some of these tensions; their status varies with the systems and assumptions examined. Nevertheless, each additional prescription makes the inferred material halo dependent on another layer of unresolved astrophysics. The point is not that every halo simulation is wrong, but that these adjustments do not by themselves establish a separately existing dark-material substance.[47, 1]

There is a deeper interpretive problem. A dynamical acceleration fixes a gravitational response, *not* a unique decomposition of that response into matter of different identities. The familiar operation of attributing the difference between a measured field and an ordinary-material calculation to an unseen halo is a model-dependent inverse inference. A gravitational equivalent mass is not an inventory of particles simply because it carries mass units. In a rotating, flattened, pressure-supported or relativistic system, the geometry and state used in the inversion matter.[4, 3, 2]

10.2 The CPTG replacement: gravitational response rather than additional material

CPTG replaces the extra material source with a geometric mechanism while retaining the Einstein tensor and the ordinary material stress–energy. Its field equation, Eq. (3), includes curvature polarization and transport. Their sum can be moved to the source side to define the *equivalent* geometric response tensor in Eq. (4); this is an identity within the coupled field theory, not a proposal to add a fluid whose particles have escaped detection. The gravitational response is organized

by the source distribution, the measured physical state and the appropriate geometry, rather than by assigning a freely invented mass inventory to the unexplained part of the field.[2, 3]

For resolved disks, the geometric contribution is expressed in a declared gravitational mass equivalent, $M^{\mathrm{eq}}_{\mathrm{grav}}(R) = R^2 g_{\mathrm{C}}(R)/G$, and compared with the independently reconstructed material inventory. The two-register accounting developed in Sections 2 and 3 also shows why the inventory remainder is *not* automatically identical to a local polarization-only force or a unique covariant mass density: flattened material geometry and the force-reference convention contribute to the comparison. The field itself is the physically primary object. A mass equivalent names what that field does gravitationally, not a missing stock of unseen matter.[4, 3]

The same distinction extends to pressure-supported galaxies and clusters through their hydrostatic or Jeans support operators, without exchanging the material tensor for a cold-dark-matter stress–energy component. At cosmic scale, CPTG also assigns accelerated evolution to its registered native curvature–transport history rather than to a separately introduced dark-energy fluid; the native, acoustic and luminosity constructions retain their distinct observational meanings. These are different reductions of the registered program, not permission to insert a new adjustable component whenever an individual object is difficult.[3, 9, 14, 15, 16]

10.3 The Decisive Difference: Fixed Laws Versus Fitted Parameters

The significance of CPTG extends beyond reproducing individual observations. It concerns the restrictions under which those results are obtained. In conventional cosmology, the ΛCDM parameters are inferred by fitting observations. CPTG instead introduces registered geometric laws whose coefficients and projection rules cannot be independently adjusted to accommodate each new observational discrepancy.

The distinction is particularly consequential in cosmology. The fixed native curvature–transport history determines the cosmic acceleration transition. Its registered acoustic projection carries the CMB boundary into the DESI distance ratios without fitting a separate DESI ruler or global normalization. The luminosity projection produces its distinct Hubble endpoint without altering the native history. The CMB comparison evaluates a fixed geometric angular envelope, while the nuclear construction applies separately registered source and transport laws, including the mass-seven optical depth $\tau_7 = \ln \pi$.[14, 16, 15, 18, 8, 20]

These calculations use their declared observational inputs, and individual gravitational reconstructions retain their measured support conditions. Those inputs do not authorize changing the underlying geometric laws whenever a comparison becomes unfavorable. A failed comparison must remain a failure under the registered construction, rather than becoming an opportunity to introduce an additional adjustable term.

This is the foundational challenge CPTG presents to the dark-matter paradigm. The question is not merely which description can reproduce a selected observation, but how much independent freedom must be introduced to do so. A constrained geometric framework capable of accounting for gravitational structure, cosmic observations, and nuclear abundances offers a fundamentally different explanatory economy from assigning additional material or vacuum-energy components to the gravitational field.

10.4 The standard that decides between the interpretations

A reckoning must concern the *full* gravitational evidence, not only cases in which a conventional halo is difficult to model. In particular, the CMB anisotropy structure, gravitational lensing and the separation between lensing-inferred gravity and X-ray gas in merging clusters are consequential empirical constraints often used to support dark matter. The Bullet Cluster is a conspicuous example. A geometric replacement must account for those measurements in its own field variables and observation operators; relabeling an inferred lensing mass as geometry is not itself an explanatory calculation. The galaxy, cluster, and cosmological constructions address distinct aspects of that task through their corresponding field equations and observation operators.[43, 49, 3, 16, 18]

The resulting scientific challenge is direct: if the observed gravitational phenomena can be generated consistently by ordinary material sources and their coupled geometric response, the postulate of an additional cold material inventory is no longer required *within that explanation*. The burden then shifts from adjusting an unseen halo to demonstrating which parts of the measured field genuinely require new material degrees of freedom. This does not amount to an experimental proof that no dark particle can exist; it identifies the material dark-matter postulate as an avoidable assumption in the

CPTG framework rather than an observational fact.[2, 4, 3]

The reckoning concerns explanatory necessity. The decisive step under examination is the promotion of a gravitational discrepancy into a second material inventory without an independently identified constituent. CPTG offers a different accounting through the completed Einstein–polarization–transport field: its additional geometric response can be quantified without assigning it to invisible material particles. In the registered reconstructions an extra cold-material source is not required as an input; the full CMB, lensing and dynamical observations remain the physical standard against which that replacement must be judged.[2, 4, 3, 7]

11 Geometric Mass and the Unified Architecture of Gravity

Gravitational mass is a measure of how a gravitational field acts, while material mass describes the matter supplying its source. CPTG gives this distinction a common physical structure. Its Einstein term, geometric polarization, and transport response determine a completed field whose observed expression depends on the matter distribution, motion, pressure, and geometric organization of each object. Across galaxies, clusters, compact states, cosmological observations and primordial nuclear reactions, the registered calculations give that framework a concrete quantitative form.[2, 3, 4, 7]

The synthesis has two complementary levels. The shared gravitational parent retains Einstein curvature and material stress–energy together with geometric polarization and transport. Each registered realization supplies its own reduced operators, state conditions, matching requirements and observation coordinates; these are not all independently derived as one interchangeable scalar field law. The nuclear vertex–bridge–closure–saturation construction instead obeys its separately defined conserved reaction-space identity, not the Einstein field equation. The preceding dark-matter reckoning concerns the necessity of an additional material hypothesis; the account here concerns the positive physical consequences across these distinct realizations.[2, 3, 7, 8]

11.1 From local organization to a unified interpretation

In disks, a field-derived spherical mass equivalent can be compared against a flattened material inventory or a separate radial-force reference; the two ledgers are connected algebraically but retain different meanings. Pressure-supported systems instead reveal gravity through gas pressure or stellar velocity moments, and their Structural Modes remain fingerprints of the completed object-specific solutions. The native cosmological branch supplies a common cosmic phase while acoustic and luminosity observations require distinct projections. In finite compact systems, the exterior gravitational charge is preserved and its conditional localized-source/geometric split is fixed through matching and angular momentum. The nuclear construction retains a separate conserved four-sector reaction-space identity. These are different observables and procedures, not independent laws selected to imitate each other.[4, 3, 14, 7, 8]

The foundational continuity ledger (Table 1) identifies the shared field architecture and each regime's observation interface. The consequences below concern what those completed geometries imply for mass, time and cosmic evolution.

11.2 Six consequences of one geometric-mass principle

The compact and cosmological constructions extend the significance of geometric response beyond conventional galactic mass accounting. Table 17 collects six physical consequences while leaving their complete derivations in their respective chapters.[7, 2, 3]

Physical question	CPTG construction and consequence
Black-hole information	A finite compact continuation removes the infinite-curvature singular endpoint as an information sink in the stationary model; the exterior horizon and semiclassical setting are retained.
Singularity and geodesic incompleteness	The registered compact profile has a regular center and continuous Schwarzschild/Kerr matching, replacing the singular classical continuation in the finite compact solution class.
Compact mass gap	The observed exterior mass remains fixed, while the matching-radius cubic maps it to $M_c = M_\infty/A_{\rm pol}$ and a different localized-source mass coordinate.
"Missing baryons" in compact mass accounting	$M_\infty = M_c + M_{\rm geom}$ distinguishes localized source mass from completed geometric gravitational charge, without reclassifying the geometric share as extra matter.
Cosmic acceleration without dark energy	The native $E_\pi(a)^2 = (1 - 1/\pi) + (1/\pi)a^{-\pi}$ history crosses from deceleration to acceleration at $z_{\rm tr} = 0.52332$ and has $q_{\pi,0} = -1/2$. Curvature transport and its geometric late-time weight replace a separately introduced cosmological-constant fluid.
Unification of cosmic scales	One source–curvature–polarization–transport architecture connects resolved disks, pressure-supported clusters, cosmological observation coordinates, and finite compact states; its nuclear source program uses a distinct conserved reaction-space realization.

Table 17: Six physical consequences emphasized by the registered CPTG constructions. Each consequence follows from its corresponding registered equations and observation operators.

Together these consequences follow the physical route from material source and dynamical state to a completed coupled field, and from that field to a clearly specified mass, time, distance, or reaction-space readout. The distinctions among those readouts strengthen the common geometric account rather than making them interchangeable.[7, 3, 21, 2]

11.3 Conclusion: gravity as an organized geometric response

The essential change in physical interpretation is that ordinary material is the source, but its mass inventory alone is not a complete description of the gravitational field. The Einstein tensor remains continuously part of the coupled theory. Polarization and transport express the field's additional geometrical organization, and compatible quantities are matched where reductions overlap. The same interpretation supports multiple, correctly distinguished readouts: galaxy field equivalents, pressure-supported dynamical fields, native cosmic projections, finite compact exterior charge and internal clocks.[2, 3, 7]

The dark-matter interpretation is examined directly in Section 10. A separate consequence is the fixed native curvature–transport history, which supplies an accelerating cosmological account without a distinct dark-energy fluid. Both have quantitative meaning through their declared equations and the appropriate observational mappings.[2, 4, 14, 16]

The unifying strength of CPTG is not agreement with observation alone, but agreement obtained under registered geometric constraints. The native cosmic history, acoustic and luminosity projections, gravitational response laws, and nuclear transport prescriptions retain their declared mathematical identities across their respective applications. Their consistency is therefore a test of the framework, not permission to retune its foundations.

The larger point is constructive rather than merely a criticism of earlier cosmology. Across rotating, pressure-supported, relativistic and cosmological states, the observed expression of gravity depends on the organization of the completed coupled field, not solely on a scalar total of material mass. Structural Mode measures that organization after the field is solved; it does not manufacture the response. *Geometric mass is the gravitational expression of that organization.*[2, 3, 5, 7]

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